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The forecast ability of a belief-based momentum indicator in full-day, daytime, and nighttime volatilities of Chinese oil futures

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ABSTRACT

China launched its domestic crude oil futures market in 2018 and this nascent market is steadily developing. Studying volatility predictability for Chinese crude oil futures has important academic and practical significance. In this study, we examine the predictive impacts of a novel predictor, the belief-based momentum indicator of conditional past return (CPR), a cluster of predictors from the domestic (Chinese) stock and oil markets and the US stock and options markets, and global finance-related indicators. In addition to the full-day oil realized volatility (ORV), we also forecast the daytime and nighttime ORVs. First, we find that the novel CPR indicator has significant incremental predictive information for the full-day, daytime, and nighttime ORVs, whether in the short, middle, or long term. Furthermore, the CPR indicator is the most reliable among the four endogenous predictors. Second, predictors of the Chinese stock market have no effective predictive information; instead, predictors from the US stock and options market have great impacts on future Chinese ORVs. Third, our evidence shows that the predictability of the night ORV is half that of the daytime ORV, and the VIX is relevant for the nighttime ORV. Our research helps Chinese policymakers improve the crude oil market mechanism and prevent market risks.

1. Introduction

Oil price uncertainty strongly affects global macroeconomics and thus has influential decision-making effects for policymakers, global companies, investors, and oil sector workers, among others. Investigating and improving the predictability of oil price uncertainty is pertinent to academics and market participants. Chinese financial markets are increasingly important to global investors, and China launched its domestic oil futures market on 26 March 2018. Many studies underscore the importance of Chinese oil futures (Wen et al., 2021; Hu and Jiang, 2023; Huang et al., 2023; Sun et al., 2023). By detecting the spillover effects among the Chinese crude oil futures market and other global crude oil futures and spot markets, Sun et al. (2023) find that the Chinese crude oil futures market is mainly a net information receiver among world crude oil markets. This indicates that China's crude oil futures market is still in the developing stage, and that market effectiveness is slightly insufficient. Therefore, studying the driving/predictive factors of China's crude oil volatility informs Chinese policymakers to formulate corresponding economic or financial policies targeting specific risk factors to improve market effectiveness. In this study, we examine the predictability of the realized volatility of Chinese crude oil futures, including the full-day oil realized volatility (ORV) and the daytime and nighttime ORVs, from the perspective of a novel belief-based momentum indicator and a cluster of domestic and foreign predictors.

There are many predictors for the realized volatility of financial assets, such as the leverage factor (Takahashi et al., 2009; Asai et al., 2012; Catania and Proietti, 2020), jump factor (Barndorff-Nielsen and Shephard, 2004), overnight factor (Liang et al., 2021), VIX, and financial stress index (Busch et al., 2011; Kambouroudis and McMillan, 2016; Gong and Lin, 2018; Gkillas et al., 2020; Li et al., 2020; Liang et al., 2020; Xiao et al., 2021). In terms of oil price uncertainty, Li et al. (2022a) examine the predictive power of multiple uncertainty indicators in forecasting monthly WTI crude oil volatility and show that the US petroleum market equity market volatility tracker index, financial stress index, and geopolitical risk index are effective predictors. However, no

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single uncertainty index performs well across all forecasting targets, e.g., both predicting short- and long-term volatilities and predicting volatilities in recessions and expansions. Das et al. (2022) also show the comovement between oil implied volatility and financial stress. Zhang et al. (2023a) propose an aligned global economic policy uncertainty (GEPU) index and show its effective predictive role in oil market volatility.

We introduce a novel belief-based momentum indicator proposed by Li et al. (2023) to study the predictability of the Chinese crude ORV. The belief-based momentum indicator is a variable that only depends on an asset's price and volume information and consists of two parts: past returns and investors' consistent beliefs. Li et al. (2022b) and Li et al. (2023) find that the effective role of the belief-based momentum indicator in asset return predictability is applicable at both the firm and market levels in the Chinese equity market. Compared to the belief-based momentum indicator in Li et al. (2022b), Li et al. (2023) release the direction limitation of investors' consistent beliefs to allow the value to be negative, which emphasizes the consistency between the direction of past returns and the direction of investors' consistent belief. The belief-based momentum indicator is a conditional past return (CPR) measurement, and we use CPR to stand for the indicator.

In this study, we mainly examine whether the CPR indicator has a predictive role on the Chinese ORV. We also consider a cluster of predictors in the literature to forecast the Chinese ORV, including endogenous predictors of the Chinese oil futures market, exogenous predictors from the Chinese equity market, the US stock and options markets, and global finance-related predictors. First, we examine which of the predictors is useful and which market has a great impact on China's ORV. Second, we show whether combining CPR and other predictors improves forecasting accuracy.

Additionally, China's crude oil futures market covers both daytime and night trading periods, where the daytime period is from 9:00 (Beijing Standard Time) to 15:00, and the nighttime period is from 21:00 on the present day to 2:30 on the next day. The opening of nighttime trading in the Chinese oil futures market is intended to include foreign investors and enhance the global influence of the market. However, many domestic investors cannot utilize the nighttime trading period well due to the limitations of their natural sleeping habits. Conversely, the daytime trading period is more convenient for Chinese investors than for foreign investors. Due to differences in domestic and foreign investors participating in the daytime and nighttime trading periods, we need to examine the potential differences in forecasting performance of the predictors for ORV during the daytime and nighttime periods. For example, do exogenous predictors from the US financial markets have a more important role in the nighttime Chinese ORV? Therefore, we decompose the full-day ORV into the daytime and nighttime ORVs and examine the predictive impacts of predictors for the full-day, daytime, and nighttime realized volatilities.

We conduct our empirical analysis using data from 26 March 2018, to 30 June 2022. The in-sample results of HAR-RV-X models comprehensively show that the CPR indicator significantly and positively impacts the one-day-ahead (short-term), five-day-ahead (middle-term), and 22-day-ahead (long-term) full-day ORVs. According to the results of the other 17 predictors, the domestic stock market has a weak impact on the oil futures market; however, the US stock and options markets have a great impact, especially the leverage factor of the US stock market, VIX, and gold implied volatility. The global financial stress index is also a source of predictive information for the Chinese ORV. Furthermore, the HAR-RV-CPR-X results show that the CPR indicator still has stable and significant coefficients among all 17 models, and each HAR-RV-CPR-X model has better in-sample predictive performance than the corresponding HAR-RV-X model. This indicates that the predictive information obtained from CPR is effective and unique and cannot be replaced by other conventional predictors. For the daytime and nighttime ORVs, we find that the CPR indicator also has significant predictive coefficients. In particular, comparing the in-sample adjusted R-squared values of predictive models in forecasting the daytime and nighttime ORVs, we find that the predictability of the nighttime ORV is almost half that of the daytime ORV.

We evaluate the out-of-sample forecasting performance of the models, and the out-of-sample forecasts range from 1 June 2020, to 30 June 2022. The CPR indicator contains effective predictive information for the full-day ORV, both for daytime and nighttime ORVs. In terms of short- and middle-term forecasting, the HAR-RV-CPR-L^{US_SM} model is the most useful for improving the predictive accuracy for full-day and daytime ORVs. The CPR indicator is the most predictive for the long-term full-day ORV, and the HAR-RV-CPR-GVX^{US} model can be used to improve the forecasting performance for the long-term daytime ORV. In terms of forecasting the nighttime ORV, the VIX has excellent performance in predicting short-, middle-, or long-term volatility, and the HAR-RV-CPR-VIX^{US} model further improves forecasting performance beyond the HAR-RV-VIX^{US} model.

Additionally, we examine the economic mechanism via liquidity, investor sentiment, or both perspectives for the predictive role of CPR on the Chinese ORV. Increasing market illiquidity results in greater market volatility due to transaction friction. Positive investor sentiment often leads to active market trading, which in turn leads to high price volatility. We find that CPR significantly and positively predicts future one- and five-day-ahead abnormal trading volume and that the contemporaneous abnormal trading volume especially impacts market volatility. Additionally, we find that CPR has no significant impact on future market illiquidity, even though contemporaneous market illiquidity is indeed positively related to oil market volatility. Therefore, our evidence supports that the predictive role of CPR on ORV is partial via market investor sentiment instead of market liquidity.

This study first contributes to the literature by introducing a belief-based momentum indicator of CPR into the oil futures markets that focuses on the predictability of ORV. The evidence supports the powerful and stable predictive role of CPR in the Chinese ORV. In addition to the role of CPR in asset price discovery in the literature (Li et al., 2022b; Li et al., 2023), our finding expands the effective role of CPR in the predictability of asset price volatility. Second, we provide novel evidence for the predictability of the daytime and nighttime ORVs in China using a cluster of predictors from the domestic and foreign financial markets. Moreover, we find evidence of greater uncertainty and less predictability of the nighttime ORV in China, which alerts the Chinese government to improve oil futures market effectiveness.

Section 2 shows the methodology and data, and our empirical results are presented in Section 3. The robustness check is conducted in Section 4. The conclusion is in Section 5.

2. Methodology and data

2.1. Oil realized volatility

Realized volatility is calculated by (Andersen et al., 2001; Andersen et al., 2003; Barndorff-Nielsen and Shephard, 2004):

$$RV_t = \sum_{j=1}^{M} r_{t,j}^2, \tag{1}$$

where $r_{t,j} = log\left(\frac{P_{t,j}}{P_{t,j-1}}\right)$ and $P_{t,j}$ is the jth price of M equal-interval sampling points for day t. Based on this calculation, the realized volatility of crude oil futures is ORV_t, which we scale by multiplying by 10,000.

2.2. Predictive models

2.2.1. HAR-RV model

We take the classic HAR-RV model proposed by Corsi (2009) as the benchmark model. Many studies forecast the natural logarithm of ORV, which is approximately Gaussian (Christiansen et al., 2012; Paye, 2012; Li et al., 2022a; Zhang et al., 2023b). We perform the same analysis.

Specifically, the HAR-RV model is expressed as:

$$\frac{1}{H}\sum_{h=1}^{H} log(ORV_{t+h}) = \beta_0 + \beta_{logORV} log(ORV_t) + \beta_{logORV^w} log\left(ORV_t^w\right) + \beta_{logORV^m} log\left(ORV_t^m\right) + \varepsilon_{t+1}^H, \quad (2)$$

where $ORV_t^w = \frac{1}{5}\sum_{h=0}^{4} ORV_{t-h}$ and $ORV_t^m = \frac{1}{22}\sum_{h=0}^{21} ORV_{t-h}$. ORV_t, ORV_t^w, and ORV_t^m are the day-, week-, and month-horizon heterogeneous components of ORV, respectively. ε_{t+1}^H is the residual term.

2.2.2. HAR-RV-X model

We introduce one predictor into the HAR-RV model to create a HAR-RV-X model,

$$\frac{1}{H}\sum_{h=1}^{H} log(ORV_{t+h}) = \beta_0 + \beta_{logORV} log(ORV_t) + \beta_{logORV^w} log\left(ORV_t^w\right) + \beta_{logORV^m} log\left(ORV_t^m\right) + \beta_x x_t + \varepsilon_{t+1}^H, \quad (3)$$

where x is one of our 18 predictors, including the CPR indicator (Li et al., 2023), leverage factors (Takahashi et al., 2009; Shephard and Sheppard, 2010; Asai et al., 2012; Catania and Proietti, 2020), jump factors (Maciel et al., 2017), overnight return information (Liang et al., 2021), VIX, oil implied volatility, gold implied volatility, global financial stress, and geopolitical risk. The descriptions of our predictors are shown in Table 1.

The calculations for some variables are shown here. Specifically, the leverage factor is $r_t^- = min(r_t, 0)$, where r_t is the daily return of the underlying asset. The jump factor is $J_t = max\{RV_t - BV_t, 0\}$, where RV_t is the realized volatility of the underlying asset, and BV_t is the realized bi-power variation (Barndorff-Nielsen and Shephard, 2004). The overnight factor is the daily overnight return of the underlying asset. The leverage and overnight factors are scaled by multiplying by 100. GPR refers to the geopolitical risk measure, which is calculated by the first-order logarithmic difference of the raw geopolitical risk index.

The belief-based momentum indicator of CPR contains two parts of information: 1) the basis of past return information and 2) investors' consistent belief information. This setting makes CPR capture a large portion of the information in investors' views about price trends so that CPR reflects the price momentum effect well (Li et al., 2022b; Li et al., 2023). The CPR indicator is calculated by

$$CPR_t = \frac{(r_t + r_{t-1} + \ldots + r_{t-21})}{22} \bullet \left[\frac{WC_t - WO_t}{WH_t - WL_t} \bullet \frac{WV_t}{max(WV_{t-i}, i = 0,1,\cdots,21)}\right], \quad (4)$$

where r_t is the asset return for day t, WC_t, WO_t, WH_t, WL_t, and WV_t are the closing, opening, highest, lowest prices, and total trading volume over the past five days, respectively.

2.2.3. HAR-RV-CPR-X model

To show the unique predictive information of CPR, we introduce one non-CPR predictor into the HAR-RV-CPR model to create a HAR-RV-CPR-X model,

$$\frac{1}{H}\sum_{h=1}^{H} log(ORV_{t+h}) = \beta_0 + \beta_{logORV} log(ORV_t) + \beta_{logORV^w} log\left(ORV_t^w\right) + \beta_{logORV^m} log\left(ORV_t^m\right) + \beta_{CPR} CPR_t + \beta_x x_t + \varepsilon_{t+1}^H, \quad (5)$$

where x is one of our 18 predictors excluding CPR.

2.3. Evaluation methods

2.3.1. In-sample estimated coefficients

The correct impact direction between a predictor and future ORV reflects a relatively effective economic implication. A significant coefficient indicates the statistical implication of the predictive role. In terms of the HAR-type models using the Ordinary Least Squared (OLS) method, a greater value in in-sample R-squared means that the corresponding predictors can better explain the changes in ORV.

2.3.2. Out-of-sample evaluation

In terms of out-of-sample forecasting analysis, the rolling window method is used to generate forecasts, and the training window is M. We use the following methods to evaluate the performance of our eight predictive models.

The out-of-sample R-square (R_{OOS}^2) of a predictive model is as follows:

Table 1
Variable description and data source.

Variable	Description	Data source
Panel A. Oil realized volatility and endogenous predictors of the Chinese oil futures market		
$logORV_t$	The logarithm of oil realized volatility	WIND Database: https://www.wind.com.cn/
L_{t-1}	The leverage factor of the Chinese oil futures market	WIND Database: https://www.wind.com.cn/
J_{t-1}	The jump factor of the Chinese oil futures market	WIND Database: https://www.wind.com.cn/
O_{t-1}	The overnight factor of the Chinese oil futures market	WIND Database: https://www.wind.com.cn/
CPR_{t-1}	Belief-based momentum indicator of the Chinese oil futures market	WIND Database: https://www.wind.com.cn/
Panel B. Exogenous predictors from the Chinese stock market		
$RV_{t-1}^{CN_SM}$	The realized volatility of the Chinese SSEC index	https://realized.oxford-man.ox.ac.uk/
$L_{t-1}^{CN_SM}$	The leverage factor of the Chinese SSEC index	https://realized.oxford-man.ox.ac.uk/
$J_{t-1}^{CN_SM}$	The jump factor of the Chinese SSEC index	https://realized.oxford-man.ox.ac.uk/
$O_{t-1}^{CN_SM}$	The overnight factor of the Chinese SSEC index	https://realized.oxford-man.ox.ac.uk/
Panel C. Exogenous predictors from the US stock and options markets		
$RV_{t-1}^{US_SM}$	The realized volatility of the US S&P 500 index	https://realized.oxford-man.ox.ac.uk/
$L_{t-1}^{US_SM}$	The leverage factor of the US S&P 500 index	https://realized.oxford-man.ox.ac.uk/
$J_{t-1}^{US_SM}$	The jump factor of the US S&P 500 index	https://realized.oxford-man.ox.ac.uk/
$O_{t-1}^{US_SM}$	The overnight factor of the US S&P 500 index	https://realized.oxford-man.ox.ac.uk/
$IDEMV_{t-1}^{US_SM}$	US's Infectious Disease Equity Market Volatility Tracker index	http://www.policyuncertainty.com/index.html
VIX_{t-1}^{US}	US's CBOE Volatility Index	FRED website: https://fred.stlouisfed.org/
OVX_{t-1}^{US}	US's CBOE Crude Oil ETF Volatility Index	FRED website: https://fred.stlouisfed.org/
GVX_{t-1}^{US}	US's CBOE Gold ETF Volatility Index	FRED website: https://fred.stlouisfed.org/
Panel D. Global finance-related indicators		
$GFSI_{t-1}$	Global Financial Stress Index	OFR website: https://www.financialresearch.gov/
GPR_{t-1}	Geopolitical Risk Index	https://www.matteoiacoviello.com/

$$R^2_{OOS} = 1 - \frac{\sum_{t=M+1}^{M+Q}(RV_t - \widehat{RV}_t)^2}{\sum_{t=M+1}^{M+Q}(RV_t - \widehat{RV}_{bench,t})^2}, \tag{6}$$

where RV_t is the actual RV value for day t and $\widehat{RV}_t$ and $\widehat{RV}_{bench,t}$ are the RV forecasts of the predictive model and the benchmark model, respectively, where the latter is HAR-RV. A positive and high R^2_{OOS} means that the predictive model of interest has smaller predictive errors. We use the adjusted mean-squared forecasting errors (MSFE) statistic and the corresponding significance proposed in Clark and West (2007) to show the significance of R^2_{OOS}.

Furthermore, the loss functions and model confidence set (MCS) test are also used to demonstrate the out-of-sample model performance. We adopt the following loss function for model evaluation:

$$MSE = \frac{1}{Q}\sum_{t=M+1}^{M+Q}(RV_t - \widehat{RV}_t)^2, \tag{7}$$

$$MAE = \frac{1}{Q}\sum_{t=M+1}^{M+Q}|RV_t - \widehat{RV}_t|, \tag{8}$$

where RV_t and $\widehat{RV}_t$ are the true and predicted values of RV, respectively, M denotes the training window length, and Q refers to the forecast number. The MCS test proposed by Hansen et al. (2011) is utilized to evaluate multiple predictive models concurrently and select a set of advantage models, and many studies use this test to evaluate the model performance (Wang et al., 2016; Luo and Chen, 2020; Clements and Preve, 2021; Golosnoy and Gribisch, 2022). We obtain the MCS p-value based on the range statistic, and 0.10 is chosen as the critical P-value (Hansen et al., 2011; Li et al., 2020; Clements and Preve, 2021; Liang et al., 2022). The maximum MCS p-value of 1.00 indicates that the corresponding model is the best among all models that are compared.

2.4. Data

Table 1 shows the descriptions and data sources of our variables. The 5-min prices and daily trading volume of the Chinese oil futures (major contracts) are obtained from the WIND database; then, we calculate the full-day, daytime, and nighttime Chinese ORVs, as well as the Chinese oil leverage, jump, overnight, and CPR factors. We show the trading time of the Chinese oil futures market and ORV decomposition in Fig. 1. The daytime ORV is calculated by the high-frequency prices during the daytime trading period from 9:00 (Beijing Standard Time) to 15:00 for each day, and the nighttime ORV is calculated by the prices during the nighttime trading period from 21:00 on the present day to 2:30 on the next day. The full-day ORV is calculated by the high-frequency prices from 9:00 on the present day to 2:30 on the next day, and it is the sum of the daytime ORV and the nighttime ORV.

China launched the oil futures market on 26 March 2018. Considering all data available, the entire sample period is from 26 March 2018, to 30 June 2022. The training window M is 500 for out-of-sample forecasting, and the out-of-sample forecasts range from 1 June 2020, to 30 June 2022. It is worth noting that due to the dual impact of the COVID-19 pandemic and external crude oil market shocks,[1] the nighttime trading of the domestic crude oil futures market was closed on 22 January 2020, and was reopened on 6 May 2020. Fig. 2 shows the time series of full-day, daytime, and nighttime ORVs. Table 2 shows the descriptive statistics of the variables. The full-day ORV_t has a mean of 4.260, a high skewness of 5.274, and a very high kurtosis of 39.847. However, the full-day $log(ORV_t)$ has a skewness close to zero and a smaller kurtosis of 0.785, which indicates that $log(ORV_t)$ is approximately Gaussian and more suitable for the OLS method. The nighttime $log(ORV_t)$ has a much higher mean than the daytime $log(ORV_t)$, which indicates that the Chinese oil price uncertainty is higher during the nighttime trading period when more foreign investors and fewer domestic investors participate. We also show the time-varying proportion of nighttime ORV in full-day ORV in Fig. 3. It is clear that nighttime ORV accounts for a significant proportion of the full-day ORV, with an average proportion of 70.17% that decreases over time. This result indicates that the domestic crude oil futures market has higher fluctuations at night than during the day and is greatly affected by external shocks. In addition, we also show the time-series changes of CPR and ORVs in Fig. 4.

3. Empirical results

3.1. In-sample estimation results

3.1.1. Predicting full-day ORV by HAR-RV-X models

We forecast the short-, middle-, and long-term full-day ORVs, which are the one-, five- and 22-day-ahead RVs. In Table 3, we show the in-sample estimations of predicting the short-term full-day ORV for the HAR-RV and 18 HAR-RV-X models. Panel A reports the HAR-RV benchmark model and HAR-RV-X models by introducing endogenous predictors, including the CPR indicator, leverage, jump, and overnight factors. CPR has a significantly positive coefficient of 0.495 in predicting the one-day-horizon full-day ORV. This positive impact is consistent with our intuition that the return predictability obtained by the momentum factor of CPR also results in volatility predictability. Furthermore, among the four endogenous variables, CPR explains more changes in future ORV with a higher in-sample adjusted R-squared value of 41.8%, which means much effective prediction information is obtained from CPR.

We then shift our attention to the spillover effect of the domestic equity market in the full-day ORV. Panel B presents the four HAR-RV-X models, where X refers to each of four exogenous predictors from the domestic equity market, including market realized volatility and the corresponding leverage, jump, and overnight factors. The leverage and overnight factors of the domestic stock market have significant impacts on the short-term full-day ORV, whereas it is less for the increases in the adjusted R-squared beyond the benchmark model of the HAR-RV model for all four HAR-RV-X models. The stock market's realized volatility and jump factor have no significant impacts. The evidence partially demonstrates that the domestic equity market has a weak spillover effect on the oil futures market.

Next, we examine the spillover effect of a major foreign financial market, the US market, in the full-day Chinese ORV. Panel C shows the results of the HAR-RV-X models by introducing eight exogenous predictors from the US stock and options markets. For the US equity market, we adopt the realized volatility, leverage, jump, and overnight factors and employ IDEMV to capture the COVID-19 risk information. For the US options market, we introduce the VIX, OVX, and GVX. The results show that the US financial markets have a key role in predicting the full-day Chinese ORV. Among the eight US variables, the leverage factor of the US stock market is the most informative, followed by the overnight factor of the US stock market. The findings indicate that the return information (mainly negative returns) of the US stock market has a significant leading role in the full-day Chinese ORV. The US panic index of the VIX is also an important predictor. Synthesizing these findings, we show that the Chinese crude oil futures market is gradually internationalizing and is affected by foreign financial markets and investors (e.g., the US stock market).

In Panel D, we present the results of the HAR-RV-X models by introducing global finance-related indicators: the Global Financial Stress Index (GFSI) and the Geopolitical Risk Index (GPR). The GFSI has a significant and positive impact on the short-term Chinese ORV, whereas the GPR information does not.

[1] An astonishing fact of the US oil futures market is that the price of WTI crude oil futures closed at a negative value of $-37.63 per barrel on April 20, 2020.

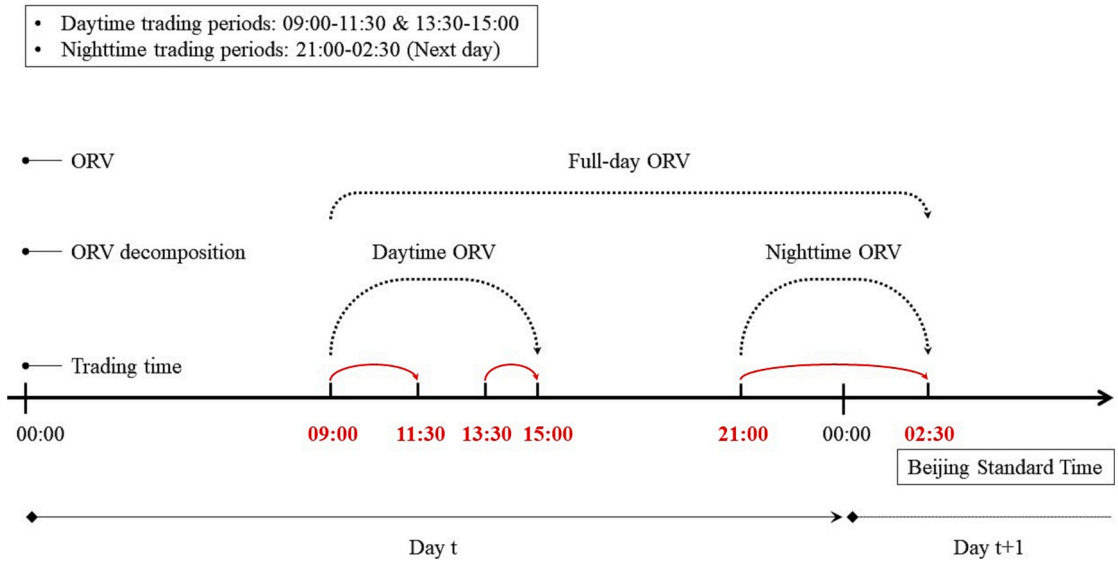

Fig 1. Trading time of the Chinese oil futures market and ORV decomposition.

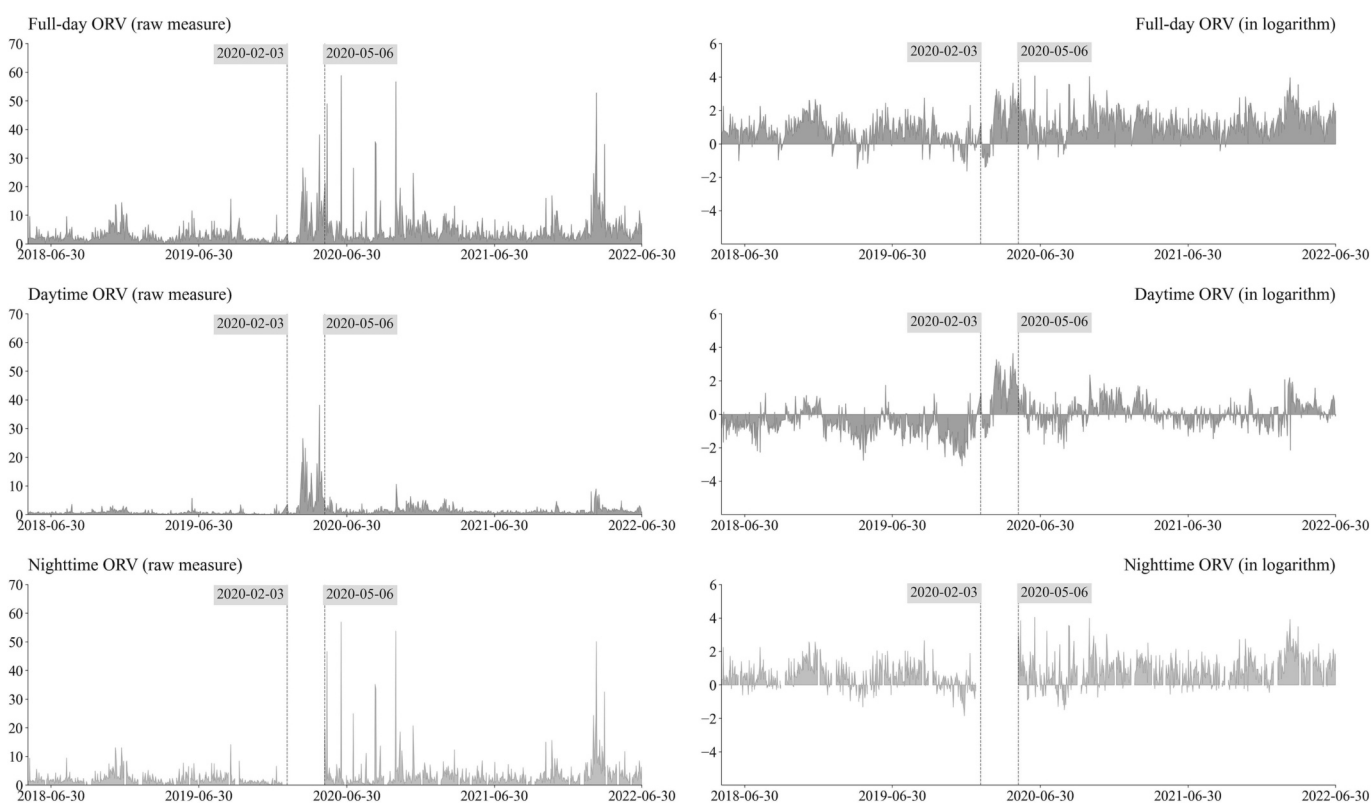

Fig 2. Full-day, daytime, and nighttime ORVs.

Table 4 reports the results of predicting the middle-term full-day ORV. In Panel A, the CPR indicator still shows a significantly positive impact, and the corresponding HAR-RV-CPR model has the highest in-sample R-squared value among the five models with endogenous factors. The results in Panel B indicate that the exogenous predictors from the domestic stock market are weak in forecasting ORV, which is consistent with the results in predicting short-term ORV. Panel C clearly shows that US financial markets have a great impact on Chinese ORV. The VIX and the leverage factor of the US stock market show better explanatory power. Similar to the previous findings in Table 3, GFSI still shows effective predictive information for the middle-term ORV, but GPR has no predictive information.

Finally, we report the estimation results for predicting the long-term full-day ORV in Table 5. These findings are in line with the results in forecasting the short-term and middle-term ORVs. In Panel A, the endogenous predictors of the leverage and overnight factors no longer have predictive information for the long-term ORV, whereas CPR still has an important predictive role. In Panel C, the gold implied volatility

Table 2
Descriptive statistics.

Variables	Mean	Std. Dev.	Skewness	Kurtosis
Panel A. Oil realized volatilities in China				
Full-dayORV_t	4.260	5.251	5.274	39.847
Full-day $logORV_t$	1.073	0.831	0.171	0.785
Daytime$logORV_t$	−0.190	0.925	0.402	1.042
Nighttime$logORV_t$	0.731	0.835	0.500	0.969
Panel B. Endogenous predictors				
L_{t-1}	−0.796	1.480	−3.064	13.216
J_{t-1}	1.258	3.523	9.237	104.390
O_{t-1}	−0.059	2.124	0.228	5.143
CPR_{t-1}	0.090	0.257	2.302	15.541
Panel C. Exogenous predictors from the Chinese stock market				
$RV_{t-1}^{CN_SM}$	19.167	19.207	3.652	19.111
$L_{t-1}^{CN_SM}$	−0.404	0.741	−3.484	18.884
$J_{t-1}^{CN_SM}$	17.604	35.614	7.149	82.538
$O_{t-1}^{CN_SM}$	−0.064	0.567	−1.144	8.061
Panel D. Exogenous predictors from the US stock and options markets				
$RV_{t-1}^{US_SM}$	26.179	72.032	9.214	104.737
$L_{t-1}^{US_SM}$	−0.411	0.927	−5.210	46.050
$J_{t-1}^{US_SM}$	21.110	104.453	19.381	459.595
$O_{t-1}^{US_SM}$	0.042	0.635	−1.726	26.940
$IDEMV_{t-1}^{US_SM}$	9.087	10.659	1.611	3.450
VIX_{t-1}^{US}	21.085	8.873	2.502	10.355
OVX_{t-1}^{US}	43.980	25.869	4.504	27.734
GVX_{t-1}^{US}	16.250	5.277	1.520	4.518
Panel E. Global finance-related indicators				
$GFSI_{t-1}$	−1.836	2.263	2.276	6.682
GPR_{t-1}	0.133	0.571	0.267	2.238

Note: The entire sample period is from 26 March 2018 to 30 June 2022.

of the US options market shows a stronger impact, and its predictive information is second only to the VIX among the eight exogenous predictors from the US financial markets.

In short, these in-sample results demonstrate that the belief-based momentum indicator of CPR has a necessary predictive role in short-, middle-, and long-term full-day ORVs. Regarding spillover effects, we show that the domestic equity market has a weak impact on the domestic oil futures market; however, the US stock and options markets have a great impact, especially the VIX, the stock market's leverage factor, and the gold implied volatility. For two global finance-related indicators, the global financial stress index shows a significant and persistent effect in predicting the full-day Chinese ORV.

3.1.2. *Predicting full-day ORV by HAR-RV-CPR-X models*

We show the results of the HAR-RV-CPR-X models with CPR and one of the other 17 predictors. Table 6 shows the estimations for predicting the short-term full-day ORV. Among all 17 models, CPR has stable and significant coefficients between the interval from 0.380 to 0.499. The adjusted R-squared of the HAR-RV-CPR-X models increases significantly beyond that of the corresponding HAR-RV-X models. This indicates that the predictive information obtained from CPR is effective and unique and cannot be replaced by other conventional predictors. Among all 17 models, the HAR-RV-CPR-L^{US_SM} model has the highest in-sample explanation power with an in-sample adjusted R-squared of 43.7%, followed by the HAR-RV-CPR-O^{US_SM} and HAR-RV-CPR-VIX^{US} models, which have adjusted R-squared values of 43.4% and 43.3%, respectively.

Tables 7 and 8 report the estimations for predicting the middle- and long-term full-day ORVs, respectively. In all HAR-RV-CPR-X models, CPR still shows significantly positive impacts. For predicting the middle-term ORV, the HAR-RV-CPR-VIX^{US} model performs best regarding in-sample prediction, followed by the HAR-RV-CPR-GVX^{US} and HAR-RV-CPR-L^{US_SM} models. For predicting the long-term ORV, the HAR-RV-CPR-VIX^{US} model performs best, followed by the HAR-RV-CPR-GVX^{US} and HAR-RV-CPR-$IDEMV^{US_SM}$ models.

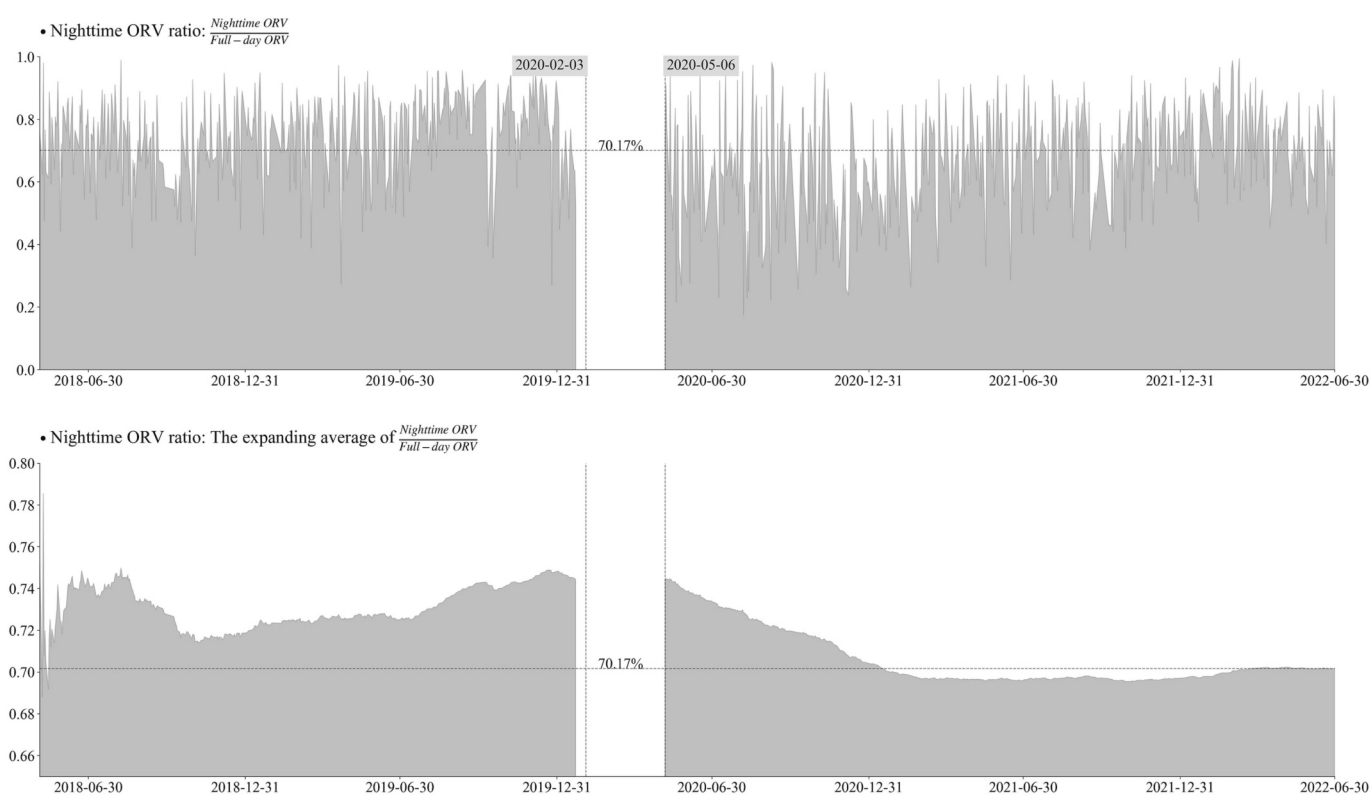

Fig 3. The proportion of nighttime ORV in full-day ORV.

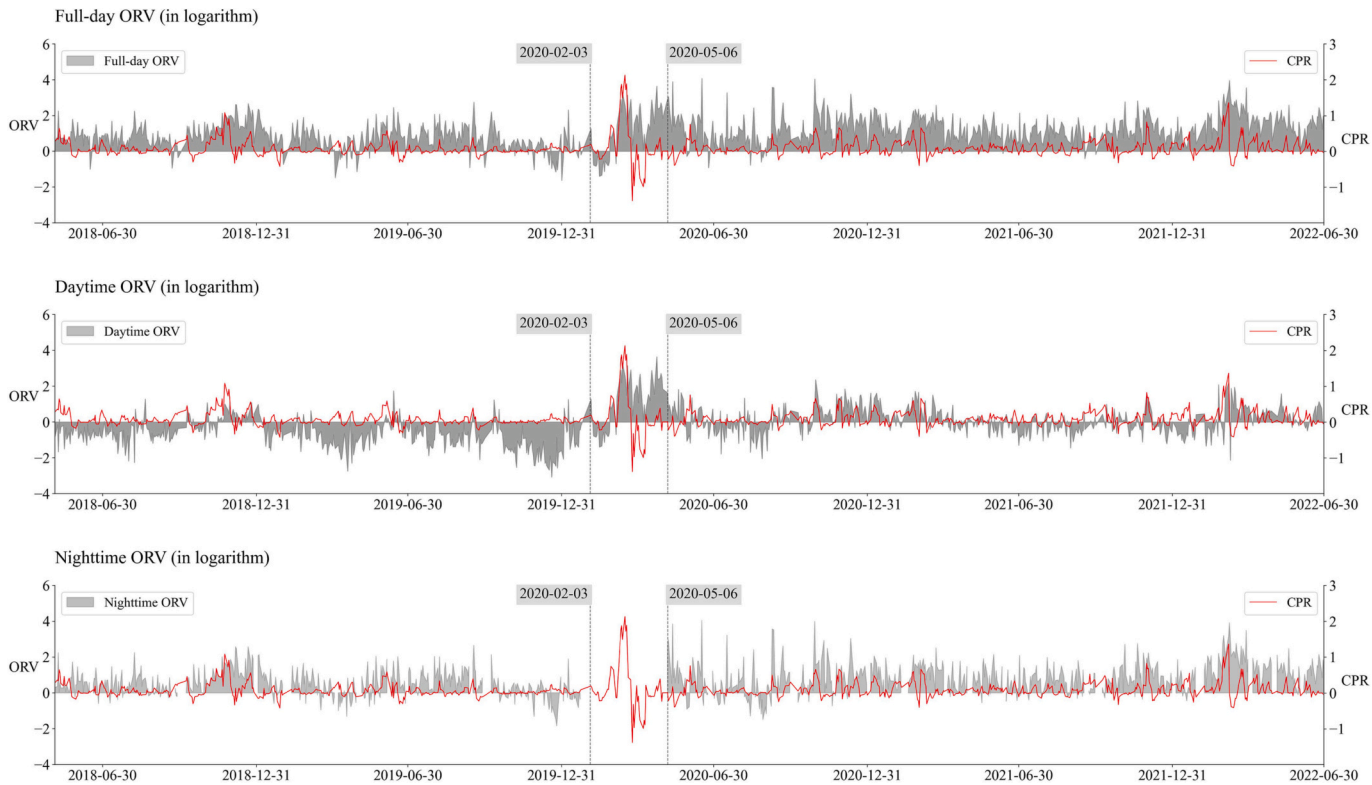

Fig 4. Time-series graphs of CPR and ORV.

3.1.3. *Predicting the daytime and nighttime ORVs*

The opening of the Chinese oil futures nighttime market is intended to allow foreign investors to trade. Therefore, investors' trading behaviour in the Chinese oil market could be different during the daytime versus nighttime periods. It is necessary to detect whether the predictive performance of potential predictors and models differs in the daytime versus nighttime ORVs.[2]

Table 9 presents the results for predicting the daytime ORV, including the coefficients and their t-statistics of the predictors of interest and the adjusted R-squared. First, the significant and positive impact of the CPR indicator exists in predicting short-, middle-, and long-term daytime ORVs. Second, the model with the endogenous leverage factor has the best in-sample predictive performance in forecasting the short-term daytime ORV, and the model with the VIX performs best in forecasting the middle- and long-term daytime ORVs. Third, the predictors obtained from the domestic stock market have no crucial impact on the daytime ORV, which is the same finding as in our previous results. However, the predictors of the US financial markets are important to predict China's daytime ORV.

The estimation results for predicting the nighttime ORV are shown in Table 10. Most of the results are consistent with the findings of predicting the daytime ORV. However, in one finding, the HAR-RV-OVX^{US} model performs best in forecasting the short-term nighttime ORV instead of the model with the endogenous leverage factor. There are more effective predictors from the US market, such as VIX^{US}, RV^{US_SM}, and L^{US_SM}, than from the domestic Chinese market. These findings demonstrate that the exogenous economic variables of foreign financial markets play a more important role in Chinese nighttime oil price volatility. This is also in line with the intuition that many domestic investors do not participate in nighttime oil trading due to their natural sleeping habits, and that foreign investors and foreign economic elements have more important impacts on the nighttime ORV.

Comparing the results of forecasting the daytime and nighttime ORVs, we find that the predictability of the nighttime ORV is less than that of the daytime ORV. The adjusted R-squared values in Table 10 are almost half those in Table 9. This reminds us that the Chinese government and academia need to focus on the influencing factors and predictability of the nighttime ORV despite the relatively high predictability of the full-day ORV.

3.2. *Out-of-sample performance*

We report the R^2_{OOS} of predictive models in forecasting the full-day ORV in Table 11. When predicting the short-term full-day ORV, the HAR-RV-L^{US_SM} model has the highest R^2_{OOS} of 4.633%, and the HAR-RV-CPR model also shows a significant R^2_{OOS} of 3.306%, which ranks third in forecasting performance among the 18 HAR-RV-X models. We then show the performance of the HAR-RV-CPR-L^{US_SM} model together with the CPR and L^{US_SM} factors and find that the model has a better predictive performance than any of the HAR-RV-X models. This indicates that the CPR indicator brings incremental predictive information and improves the forecasting performance in the full-day ORV. When predicting the middle-term full-day ORV, the HAR-RV-L^{US_SM} model also performs best, followed by the HAR-RV-CPR model, with a significant R^2_{OOS} of 5.008%. The HAR-RV-CPR-L^{US_SM} model together with CPR and the best predictor performs better than the HAR-RV-L^{US_SM} and HAR-RV-CPR models.

For the long-term full-day ORV, the HAR-RV-CPR model has a significant R^2_{OOS} of 2.638%, which is the highest among the HAR-RV-X models. However, many remaining models show small and negative R^2_{OOS}. These findings indicate that the effective predictive information in forecasting the long-term full-day ORV is less from the out-of-sample perspective, and that the CPR indicator is instead useful in this case.

[2] All predictors excluding the endogenous jump factor are the same. The endogenous jump factor in predicting the daytime (nighttime) ORV is calculated based on the daytime (nighttime) ORV series instead of the original ORV series.

Table 3
Estimation results of HAR-RV-X models for predicting the one-day horizon full-day oil realized volatility

Models	Intercept	t-stat	β_{logORV}	t-stat	β_{logORV^w}	t-stat	β_{logORV^m}	t-stat	β_x	t-stat	AdjustR^2
Panel A. The HAR-RV benchmark model and HAR-RV-X models by introducing endogenous predictors											
HAR-RV	0.167***	3.661	0.200***	5.211	0.500***	7.804	0.148**	2.446			0.397
HAR-RV-CPR	0.118***	2.593	0.155***	4.043	0.475***	7.533	0.221***	3.654	0.495***	6.052	0.418
HAR-RV-L	0.131***	2.878	0.132***	3.300	0.522***	8.231	0.170***	2.853	−0.077***	−5.299	0.413
HAR-RV-J	0.164***	3.610	0.260***	6.010	0.481***	7.501	0.133**	2.199	−0.020***	−2.963	0.402
HAR-RV-O	0.164***	3.621	0.187***	4.890	0.510***	7.996	0.154**	2.567	0.033***	3.402	0.404
Panel B. HAR-RV-X models by introducing exogenous predictors from the Chinese stock market into the HAR-RV model											
HAR-RV-RV^{CN_SM}	0.129**	2.530	0.195***	5.089	0.486***	7.509	0.170***	2.749	0.002	1.633	0.398
HAR-RV-L^{CN_SM}	0.133***	2.820	0.185***	4.798	0.498***	7.790	0.168***	2.770	−0.076***	−2.719	0.401
HAR-RV-J^{CN_SM}	0.163***	3.452	0.198***	5.163	0.498***	7.752	0.151**	2.471	0.000	0.359	0.397
HAR-RV-O^{CN_SM}	0.158***	3.456	0.186***	4.824	0.511***	7.976	0.154**	2.551	−0.093**	−2.555	0.401
Panel C. HAR-RV-X models by introducing exogenous predictors from the US stock and options markets into the HAR-RV model											
HAR-RV-RV^{US_SM}	0.129***	2.800	0.187***	4.910	0.413***	6.238	0.250***	3.913	0.001***	4.590	0.409
HAR-RV-L^{US_SM}	0.107**	2.359	0.166***	4.414	0.460***	7.322	0.217***	3.638	−0.158***	−7.123	0.426
HAR-RV-J^{US_SM}	0.152***	3.338	0.193***	5.064	0.477***	7.423	0.179***	2.926	0.001***	2.949	0.402
HAR-RV-O^{US_SM}	0.151***	3.370	0.174***	4.620	0.520***	8.264	0.177***	2.980	−0.207***	−6.511	0.421
HAR-RV-IDEMV^{US_SM}	0.151***	3.326	0.195***	5.115	0.456***	7.065	0.141**	2.348	0.008***	4.060	0.406
HAR-RV-VIX^{US}	−0.074	−1.245	0.179***	4.737	0.390***	5.963	0.184***	3.093	0.016***	6.237	0.419
HAR-RV-OVX^{US}	0.070	1.397	0.190***	4.993	0.430***	6.592	0.149**	2.497	0.004***	4.639	0.409
HAR-RV-GVX^{US}	−0.065	−0.937	0.193***	5.080	0.424***	6.442	0.142**	2.373	0.020***	4.452	0.408
Panel D. HAR-RV-X models by introducing global finance-related indicators into the HAR-RV model											
HAR-RV-GFSI	0.312***	5.403	0.191***	5.016	0.440***	6.742	0.151**	2.514	0.041***	4.043	0.406
HAR-RV-GPR	0.167***	3.629	0.200***	5.208	0.500***	7.798	0.148**	2.443	0.002	0.051	0.397

Note: The entire sample period is from 26 March 2018, to 30 June 2022. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Additionally, the HAR-RV-CPR-O^{US_SM} model together with the CPR and the best predictor shows a higher R^2_{OOS} of 2.847%. It is worth noting that the VIX predictor has less effective information, which is counter to the previous in-sample findings shown in Tables 3, 4, and 5. Considering the in-sample period from 26 March 2018 to 30 June 2022 and the out-of-sample period from 1 June 2020 to 30 June 2022, the potential reason for the failure of the VIX in the out-of-sample forecasting is that the VIX has a time-varying predictive role in the long-term full-day ORV. Another reason may be that the VIX asymmetrically affects the long-term daytime and nighttime ORVs. We show the asymmetrical role in the following results for predicting the daytime and nighttime ORVs. Even when performing worse in the long-term full-day ORV, the VIX still shows an effective predictive role in the short-term and middle-term full-day ORVs.

We present the out-of-sample results for predicting the daytime and nighttime ORVs in Tables 12 and 13, respectively. For the daytime ORV, among all HAR-RV-X models, the performance of the model with CPR ranks first, second, and second in predicting short-, middle-, and long-term daytime ORVs, respectively. The leverage factor of the US stock market is another key predictor for the short- and middle-term daytime ORVs, and the GVX is the most predictive variable for the long-term daytime ORV. The VIX fails to effectively predict the long-term

Table 4
Estimation results of HAR-RV-X models for predicting the five-day horizon full-day oil realized volatility

Models	Intercept	t-stat	β_{logORV}	t-stat	β_{logORV^w}	t-stat	β_{logORV^m}	t-stat	β_x	t-stat	AdjustR^2
Panel A. The HAR-RV benchmark model and HAR-RV-X models by introducing endogenous predictors											
HAR-RV	0.245***	7.724	0.148***	5.544	0.460***	10.262	0.167***	3.971			0.524
HAR-RV-CPR	0.204***	6.502	0.110***	4.145	0.437***	10.005	0.231***	5.532	0.429***	7.575	0.549
HAR-RV-L	0.219***	6.909	0.098***	3.525	0.475***	10.751	0.184***	4.429	−0.057***	−5.614	0.538
HAR-RV-J	0.243***	7.689	0.199***	6.590	0.444***	9.921	0.155***	3.681	−0.017***	−3.549	0.529
HAR-RV-O	0.243***	7.714	0.137***	5.143	0.469***	10.561	0.174***	4.164	0.031***	4.622	0.533
Panel B. HAR-RV-X models by introducing exogenous predictors from the Chinese stock market into the HAR-RV model											
HAR-RV-RV^{CN_SM}	0.228***	6.393	0.146***	5.453	0.453***	10.013	0.177***	4.112	0.001	1.082	0.524
HAR-RV-L^{CN_SM}	0.231***	7.016	0.142***	5.261	0.459***	10.245	0.176***	4.147	−0.032	−1.636	0.524
HAR-RV-J^{CN_SM}	0.242***	7.367	0.147***	5.491	0.458***	10.198	0.170***	3.985	0.000	0.412	0.523
HAR-RV-O^{CN_SM}	0.239***	7.520	0.139***	5.158	0.467***	10.430	0.171***	4.078	−0.063**	−2.497	0.526
Panel C. HAR-RV-X models by introducing exogenous predictors from the US stock and options markets into the HAR-RV model											
HAR-RV-RV^{US_SM}	0.211***	6.641	0.137***	5.178	0.381***	8.300	0.259***	5.857	0.001***	5.940	0.539
HAR-RV-L^{US_SM}	0.195***	6.253	0.120***	4.601	0.428***	9.862	0.224***	5.440	−0.130***	−8.495	0.555
HAR-RV-J^{US_SM}	0.231***	7.286	0.142***	5.350	0.437***	9.770	0.198***	4.680	0.001***	4.271	0.532
HAR-RV-O^{US_SM}	0.235***	7.517	0.132***	5.002	0.471***	10.688	0.186***	4.471	−0.129***	−5.825	0.539
HAR-RV-IDEMV^{US_SM}	0.228***	7.281	0.142***	5.422	0.412***	9.254	0.160***	3.868	0.009***	6.391	0.542
HAR-RV-VIX^{US}	−0.025	−0.618	0.124***	4.870	0.337***	7.620	0.208***	5.179	0.018***	10.359	0.570
HAR-RV-OVX^{US}	0.154***	4.473	0.139***	5.274	0.394***	8.712	0.169***	4.093	0.004***	6.297	0.541
HAR-RV-GVX^{US}	−0.038	−0.803	0.140***	5.377	0.367***	8.147	0.161***	3.932	0.025***	7.964	0.552
Panel D. HAR-RV-X models by introducing global finance-related indicators into the HAR-RV model											
HAR-RV-GFSI	0.400***	10.040	0.139***	5.284	0.397***	8.811	0.170***	4.107	0.044***	6.234	0.541
HAR-RV-GPR	0.244***	7.618	0.148***	5.541	0.460***	10.267	0.167***	3.958	0.012	0.483	0.523

Table 5
Estimation results of HAR-RV-X models for predicting the 22-day horizon full-day oil realized volatility

Models	Intercept	t-stat	β_{logORV}	t-stat	β_{logORV^w}	t-stat	β_{logORV^m}	t-stat	β_x	t-stat	AdjustR^2
Panel A. The HAR-RV benchmark model and HAR-RV-X models by introducing endogenous predictors											
HAR-RV	0.455***	15.019	0.089***	3.496	0.260***	6.062	0.232***	5.755			0.385
HAR-RV-CPR	0.425***	14.053	0.061**	2.373	0.242***	5.719	0.280***	6.909	0.311***	5.664	0.405
HAR-RV-L	0.441***	14.451	0.064**	2.381	0.268***	6.264	0.240***	5.974	−0.029***	−2.928	0.39
HAR-RV-J	0.454***	14.991	0.114***	3.930	0.252***	5.857	0.226***	5.596	−0.008*	−1.789	0.387
HAR-RV-O	0.454***	14.996	0.087***	3.373	0.262***	6.112	0.233***	5.795	0.008	1.247	0.386
Panel B. HAR-RV-X models by introducing exogenous predictors from the Chinese stock market into the HAR-RV model											
HAR-RV-RV^{CN_SM}	0.439***	12.872	0.088***	3.415	0.253***	5.858	0.241***	5.841	0.001	1.036	0.385
HAR-RV-L^{CN_SM}	0.441***	14.035	0.083***	3.217	0.259***	6.045	0.240***	5.929	−0.032*	−1.730	0.387
HAR-RV-J^{CN_SM}	0.447***	14.283	0.087***	3.404	0.257***	5.973	0.238***	5.838	0.000	0.986	0.385
HAR-RV-O^{CN_SM}	0.451***	14.860	0.084***	3.247	0.264***	6.156	0.234***	5.817	−0.038	−1.559	0.386
Panel C. HAR-RV-X models by introducing exogenous predictors from the US stock and options markets into the HAR-RV model											
HAR-RV-RV^{US_SM}	0.415***	13.809	0.076***	3.055	0.170***	3.892	0.337***	8.031	0.001***	7.081	0.415
HAR-RV-L^{US_SM}	0.414***	13.735	0.067***	2.680	0.231***	5.497	0.281***	7.059	−0.107***	−7.114	0.415
HAR-RV-J^{US_SM}	0.444***	14.636	0.085***	3.336	0.242***	5.626	0.256***	6.300	0.000***	3.417	0.392
HAR-RV-O^{US_SM}	0.449***	14.891	0.080***	3.136	0.266***	6.242	0.244***	6.070	−0.078***	−3.627	0.393
HAR-RV-IDEMV^{US_SM}	0.423***	14.790	0.080***	3.312	0.182***	4.450	0.222***	5.873	0.015***	11.424	0.458
HAR-RV-VIX^{US}	0.103***	2.809	0.060**	2.577	0.098**	2.428	0.291***	7.936	0.023***	14.601	0.497
HAR-RV-OVX^{US}	0.349***	10.718	0.078***	3.149	0.184***	4.286	0.236***	6.026	0.004***	7.586	0.419
HAR-RV-GVX^{US}	0.052	1.203	0.078***	3.268	0.127***	3.070	0.226***	6.033	0.034***	12.195	0.467
Panel D. HAR-RV-X models by introducing global finance-related indicators into the HAR-RV model											
HAR-RV-GFSI	0.637***	16.901	0.080***	3.201	0.184***	4.297	0.238***	6.097	0.052***	7.724	0.421
HAR-RV-GPR	0.452***	14.820	0.089***	3.496	0.261***	6.082	0.231***	5.735	0.021	0.872	0.385

Table 6
Estimation results of HAR-RV-CPR-X models for predicting the one-day-horizon full-day ORV

Models	β_{CPR}	t-stat	β_x	t-stat	AdjustR^2
Panel A. CPR and endogenous predictors					
HAR-RV-CPR-L	0.462***	5.702	−0.071***	−4.899	0.431
HAR-RV-CPR-J	0.488***	5.984	−0.019***	−2.829	0.422
HAR-RV-CPR-O	0.499***	6.135	0.033***	3.548	0.425
Panel B. CPR and exogenous predictors from the Chinese stock market					
HAR-RV-CPR-RV^{CN_SM}	0.488***	5.959	0.001	1.272	0.418
HAR-RV-CPR-L^{CN_SM}	0.479***	5.845	−0.062**	−2.243	0.420
HAR-RV-CPR-J^{CN_SM}	0.494***	6.039	0.000	0.120	0.417
HAR-RV-CPR-O^{CN_SM}	0.486***	5.949	−0.083**	−2.314	0.420
Panel C. CPR and exogenous predictors from the US stock and options markets					
HAR-RV-CPR-RV^{US_SM}	0.414***	4.776	0.001***	2.733	0.422
HAR-RV-CPR-L^{US_SM}	0.380***	4.598	−0.133***	−5.919	0.437
HAR-RV-CPR-J^{US_SM}	0.464***	5.552	0.000*	1.755	0.419
HAR-RV-CPR-O^{US_SM}	0.405***	4.928	−0.175***	−5.475	0.434
HAR-RV-CPR-IDEMV^{US_SM}	0.468***	5.740	0.007***	3.592	0.425
HAR-RV-CPR-VIX^{US}	0.417***	5.084	0.014***	5.301	0.433
HAR-RV-CPR-OVX^{US}	0.484***	5.977	0.004***	4.543	0.429
HAR-RV-CPR-GVX^{US}	0.466***	5.718	0.018***	3.995	0.426
Panel D. CPR and global finance-related indicators					
HAR-RV-CPR-GFSI	0.479***	5.891	0.038***	3.806	0.426
HAR-RV-CPR-GPR	0.496***	6.053	−0.008	−0.233	0.417

daytime ORV in out-of-sample forecasting, which results in poor performance in predicting the full-day ORV, in line with the previous finding. Furthermore, considering the CPR and the best predictor to construct the HAR-RV-CPR-X models, we show that the HAR-RV-CPR-L^{US_SM} has an incremental predictive performance in the short- and middle-term daytime ORVs beyond all HAR-RV-X models and the HAR-RV-CPR-GVX^{US} model also has better predictive performance than the HAR-RV-X models.

For the nighttime ORV, the CPR information is helpful to significantly improve the performance in short-, middle-, and long-term nighttime ORVs, where the out-of-sample R-squared values are 1.812%, 2.597%, and 4.868%, respectively. Among the HAR-RV-X models, the model with CPR ranks fifth, fifth, and second in predictive performance, respectively. In particular, the VIX is very useful for predicting the nighttime ORV, whether short-, middle-, or long-term. Specifically, the HAR-RV-VIX^{US} model performs best among all HAR-RV-X models and has significant R^2_{OOS} values of 3.878%, 8.806%, and 6.459% for one-, five-, and 22-day-horizon nighttime ORVs, respectively. This finding further shows that exogenous economic/financial variables from the US market, especially the VIX, are influential on Chinese oil price volatility during the nighttime trading period. Furthermore, the better performance of the HAR-RV-CPR-VIX^{US} model

Table 7
Estimation results of HAR-RV-CPR-X models for predicting the five-day-horizon full-day ORV

Models	β_{CPR}	t-stat	β_x	t-stat	AdjustR^2
Panel A. CPR and endogenous predictors					
HAR-RV-CPR-L	0.405***	7.221	−0.051***	−5.141	0.560
HAR-RV-CPR-J	0.423***	7.505	−0.016***	−3.409	0.554
HAR-RV-CPR-O	0.432***	7.717	0.031***	4.846	0.559
Panel B. CPR and exogenous predictors from the Chinese stock market					
HAR-RV-CPR-RV^{CN_SM}	0.426***	7.516	0.000	0.608	0.549
HAR-RV-CPR-L^{CN_SM}	0.424***	7.456	−0.019	−1.016	0.549
HAR-RV-CPR-J^{CN_SM}	0.428***	7.561	0.000	0.107	0.549
HAR-RV-CPR-O^{CN_SM}	0.423***	7.478	−0.054**	−2.209	0.551
Panel C. CPR and exogenous predictors from the US stock and options markets					
HAR-RV-CPR-RV^{US_SM}	0.354***	5.910	0.001***	3.649	0.555
HAR-RV-CPR-L^{US_SM}	0.334***	5.861	−0.108***	−6.985	0.570
HAR-RV-CPR-J^{US_SM}	0.394***	6.829	0.000***	2.814	0.552
HAR-RV-CPR-O^{US_SM}	0.377***	6.597	−0.100***	−4.516	0.558
HAR-RV-CPR-IDEMV^{US_SM}	0.398***	7.119	0.008***	5.853	0.564
HAR-RV-CPR-VIX^{US}	0.335***	6.057	0.016***	9.250	0.585
HAR-RV-CPR-OVX^{US}	0.418***	7.526	0.004***	6.241	0.566
HAR-RV-CPR-GVX^{US}	0.390***	7.057	0.023***	7.469	0.573
Panel D. CPR and global finance-related indicators					
HAR-RV-CPR-GFSI	0.410***	7.360	0.041***	5.977	0.564
HAR-RV-CPR-GPR	0.428***	7.557	0.004	0.174	0.549

Table 8
Estimation results of HAR-RV-CPR-X models for predicting the 22-day-horizon full-day ORV

Models	β_{CPR}	t-stat	β_x	t-stat	AdjustR^2
Panel A. CPR and endogenous predictors					
HAR-RV-CPR-L	0.299***	5.430	−0.024**	−2.462	0.408
HAR-RV-CPR-J	0.308***	5.609	−0.007	−1.621	0.406
HAR-RV-CPR-O	0.312***	5.674	0.008	1.302	0.405
Panel B. CPR and exogenous predictors from the Chinese stock market					
HAR-RV-CPR-RV^{CN_SM}	0.309***	5.602	0.000	0.650	0.404
HAR-RV-CPR-L^{CN_SM}	0.305***	5.527	−0.023	−1.236	0.405
HAR-RV-CPR-J^{CN_SM}	0.309***	5.624	0.000	0.747	0.404
HAR-RV-CPR-O^{CN_SM}	0.308***	5.590	−0.031	−1.284	0.405
Panel C. CPR and exogenous predictors from the US stock and options markets					
HAR-RV-CPR-RV^{US_SM}	0.202***	3.493	0.001***	5.466	0.422
HAR-RV-CPR-L^{US_SM}	0.231***	4.152	−0.091***	−5.955	0.425
HAR-RV-CPR-J^{US_SM}	0.284***	5.052	0.000**	2.295	0.407
HAR-RV-CPR-O^{US_SM}	0.282***	5.040	−0.056**	−2.571	0.408
HAR-RV-CPR-IDEMV^{US_SM}	0.256***	4.910	0.014***	11.029	0.471
HAR-RV-CPR-VIX^{US}	0.177***	3.462	0.022***	13.756	0.502
HAR-RV-CPR-OVX^{US}	0.299***	5.599	0.004***	7.535	0.437
HAR-RV-CPR-GVX^{US}	0.253***	4.903	0.033***	11.816	0.480
Panel D. CPR and global finance-related indicators					
HAR-RV-CPR-GFSI	0.286***	5.336	0.049***	7.478	0.437
HAR-RV-CPR-GPR	0.310***	5.629	0.015	0.641	0.404

compared to the HAR-RV-VIX^{US} model indicates that harnessing the predictive information of CPR and VIX can improve forecasting accuracy of nighttime ORV.

Why is the VIX so influential for the nighttime ORV in China? The VIX is a panic indicator that has a significant impact on US financial markets as well as global financial markets. Moreover, the far-reaching impacts of the COVID-19 pandemic have had damaging effects on global financial markets. The COVID-19 pandemic significantly raised the VIX, resulting in spillover effects among global financial markets, including in the Chinese financial markets. The Chinese oil futures market is affected much more during nighttime trading hours due to the greater presence of foreign investors relative to domestic investors.

In short, CPR contains effective predictive information for the Chinese ORV, both for the daytime and nighttime ORVs. In terms of the short- and middle-term oil price volatilities, the HAR-RV-CPR-L^{US_SM} model is useful for improving the predictive accuracy for the full-day ORV and the daytime ORV. In terms of the long-term oil price volatilities, few predictors, excluding CPR, have stable forecasting performance, and candidate predictors/models differ for the full-day, daytime, and nighttime ORVs. The HAR-RV-CPR model performs best for the

Table 9
Estimation results for predicting the daytime oil realized volatility

Models	one-day-horizon daytime realized volatility			five-day-horizon daytime realized volatility			22-day-horizon daytime realized volatility		
	β_x	t-stat	AdjustR^2	β_x	t-stat	AdjustR^2	β_x	t-stat	AdjustR^2
HAR-RV	–	–	0.557	–	–	0.665	–	–	0.544
HAR-RV-CPR	0.542***	6.801	0.576	0.455***	7.725	0.684	0.341***	5.429	0.557
HAR-RV-L	−0.106***	−8.054	0.584	−0.072***	−7.382	0.682	−0.036***	−3.412	0.549
HAR-RV-J	0.041	1.273	0.557	0.034	1.432	0.665	0.006	0.254	0.543
HAR-RV-O	0.035***	3.862	0.563	0.027***	4.017	0.670	0.007	0.993	0.544
HAR-RV-RV^{CN_SM}	0.000	0.012	0.557	0.001	0.978	0.665	0.001	1.434	0.544
HAR-RV-L^{CN_SM}	−0.052*	−1.938	0.558	−0.020	−1.010	0.665	−0.016	−0.787	0.543
HAR-RV-J^{CN_SM}	0.000	0.069	0.557	0.000	0.434	0.665	0.001	1.398	0.544
HAR-RV-O^{CN_SM}	−0.067*	−1.929	0.558	−0.058**	−2.256	0.666	−0.042	−1.564	0.544
HAR-RV-RV^{US_SM}	0.001***	4.893	0.567	0.001***	6.604	0.679	0.002***	7.707	0.570
HAR-RV-L^{US_SM}	−0.147***	−6.927	0.577	−0.133***	−8.554	0.688	−0.109***	−6.512	0.563
HAR-RV-J^{US_SM}	0.001***	3.503	0.562	0.001***	4.745	0.672	0.001***	4.033	0.551
HAR-RV-O^{US_SM}	−0.167***	−5.515	0.57	−0.130***	−5.785	0.676	−0.093***	−3.879	0.550
HAR-RV-IDEMV^{US_SM}	0.008***	3.324	0.561	0.011***	6.090	0.677	0.018***	10.501	0.590
HAR-RV-VIX^{US}	0.021***	7.114	0.578	0.024***	11.632	0.705	0.029***	13.455	0.616
HAR-RV-OVX^{US}	0.005***	5.396	0.569	0.004***	5.674	0.675	0.003***	3.831	0.550
HAR-RV-GVX^{US}	0.024***	4.780	0.567	0.030***	8.230	0.686	0.039***	10.208	0.588
HAR-RV-GFSI	0.039***	3.737	0.563	0.041***	5.364	0.674	0.035***	4.319	0.552
HAR-RV-GPR	−0.039	−1.122	0.557	−0.006	−0.232	0.665	−0.004	−0.150	0.543

Table 10
Estimation results for predicting the nighttime oil realized volatility

Models	one-day-horizon nighttime realized volatility			five-day-horizon nighttime realized volatility			22-day-horizon nighttime realized volatility		
	β_x	t-stat	AdjustR^2	β_x	t-stat	AdjustR^2	β_x	t-stat	AdjustR^2
HAR-RV	–	–	0.266	–	–	0.391	–	–	0.311
HAR-RV-CPR	0.435***	3.683	0.276	0.455***	5.925	0.413	0.466***	7.114	0.348
HAR-RV-L	−0.069***	−3.818	0.277	−0.045***	−3.765	0.400	−0.014	−1.383	0.312
HAR-RV-J	−0.024***	−2.984	0.273	−0.018***	−3.482	0.399	−0.009**	−2.019	0.313
HAR-RV-O	0.043***	3.760	0.277	0.032***	4.243	0.402	0.011*	1.752	0.313
HAR-RV-RV^{CN_SM}	0.002	1.645	0.268	0.001	1.127	0.391	0.000	−0.235	0.310
HAR-RV-L^{CN_SM}	−0.049	−1.540	0.268	−0.032	−1.523	0.392	−0.024	−1.366	0.312
HAR-RV-J^{CN_SM}	0.000	0.105	0.266	0.000	0.306	0.391	0.000	−0.036	0.310
HAR-RV-O^{CN_SM}	−0.073	−1.646	0.268	−0.080***	−2.777	0.396	−0.039	−1.565	0.312
HAR-RV-RV^{US_SM}	0.006***	5.021	0.285	0.006***	7.791	0.429	0.004***	6.383	0.341
HAR-RV-L^{US_SM}	−0.158***	−4.528	0.282	−0.137***	−6.051	0.414	−0.094***	−4.658	0.327
HAR-RV-J^{US_SM}	0.001	1.422	0.267	0.001**	2.382	0.394	0.001*	1.752	0.313
HAR-RV-O^{US_SM}	−0.171***	−3.370	0.275	−0.118***	−3.551	0.399	−0.076***	−2.617	0.316
HAR-RV-IDEMV^{US_SM}	0.009***	3.251	0.274	0.006***	3.351	0.398	0.008***	5.145	0.331
HAR-RV-VIX^{US}	0.023***	5.237	0.287	0.023***	8.328	0.434	0.021***	8.692	0.365
HAR-RV-OVX^{US}	0.013***	5.704	0.291	0.011***	7.176	0.423	0.005***	3.778	0.321
HAR-RV-GVX^{US}	0.019***	3.097	0.273	0.019***	4.830	0.406	0.020***	5.996	0.338
HAR-RV-GFSI	0.060***	3.508	0.275	0.050***	4.488	0.404	0.020**	1.969	0.313
HAR-RV-GPR	0.016	0.397	0.266	0.023	0.859	0.391	0.028	1.225	0.311

long-term full-day ORV among all the HAR-RV-X models mentioned. The GVX^{US} predictors are a priority for the long-term daytime ORV. In terms of forecasting the nighttime ORV, the VIX has excellent performance in terms of short-, middle-, and long-term volatility, and the HAR-RV-CPR-VIX^{US} model further improves the forecasting performance.

4. Economic mechanism and robustness check

4.1. Economic mechanism for the predictive role of CPR on the Chinese ORV: Liquidity, investor sentiment, or both perspectives?

We examine the economic mechanism via liquidity, investor sentiment, or both perspectives for the predictive role of CPR on the Chinese ORV. Increasing market illiquidity results in greater market volatility due to transaction friction. Positive investor sentiment often leads to active market trading, which in turn leads to high price volatility. Thus, from these two perspectives, we examine whether one is the main driver for the predictive role of CPR or both. To do this, we use CPR to forecast future market illiquidity and investor sentiment and examine the direct transmission mechanism of the impact of market illiquidity or investor sentiment on market volatility.

The regression results of the economic mechanism via market investor sentiment are shown in Table 14. $ABVol_t$ is the abnormal trading volume at day t, calculated by the present trading volume divided by the average daily volume over the past 22 days. Abnormal trading volume is treated as a sentiment measure in many studies (Liu

Table 11
Out-of-sample forecasting performance for predicting the full-day oil realized volatility: Out-of-sample R-squared results

Models	one-day-horizon oil realized volatility			five-day-horizon oil realized volatility			22-day-horizon oil realized volatility		
	R^2_{oos}	CW-stat	R^2_{oos} rank	R^2_{oos}	CW-stat	R^2_{oos} rank	R^2_{oos}	CW-stat	R^2_{oos} rank
HAR-RV-CPR	3.306%***	3.746	3	5.008%***	4.441	3	2.638%***	4.638	1
HAR-RV-L	0.527%***	2.753	11	−1.315%***	2.508	17	−5.531%	−0.339	16
HAR-RV-J	0.668%*	1.539	10	1.413%**	2.060	7	−0.835%	0.104	10
HAR-RV-O	1.344%***	2.840	7	0.651%***	2.489	10	−0.691%	0.070	8
HAR-RV-RV^{CN_SM}	0.430%**	1.717	14	0.392%**	2.249	11	−0.704%	0.227	9
HAR-RV-L^{CN_SM}	−0.213%	0.992	17	0.190%	1.081	12	0.477%**	1.840	3
HAR-RV-J^{CN_SM}	−0.131%	−0.890	16	−0.131%	−0.697	14	−1.524%	−2.336	11
HAR-RV-O^{CN_SM}	0.476%**	2.077	13	1.103%***	2.695	8	−0.086%	0.517	5
HAR-RV-RV^{US_SM}	1.842%***	3.538	5	2.404%***	4.205	6	−4.963%	−0.559	15
HAR-RV-L^{US_SM}	4.633%***	4.208	1	5.953%***	4.840	1	−0.165%**	2.165	6
HAR-RV-J^{US_SM}	0.490%*	1.462	12	1.019%***	3.005	9	−0.288%	0.019	7
HAR-RV-O^{US_SM}	3.793%***	3.584	2	3.997%***	4.396	4	0.799%**	2.077	2
HAR-RV-IDEMV^{US_SM}	0.987%**	1.932	8	−0.083%***	2.465	13	−2.724%***	3.824	14
HAR-RV-VIX^{US}	3.229%***	3.656	4	5.142%***	4.449	2	−2.337%***	3.668	13
HAR-RV-OVX^{US}	0.788%**	2.130	9	−0.358%*	1.558	16	−16.151%	−5.700	17
HAR-RV-GVX^{US}	1.708%***	2.922	6	3.051%***	3.761	5	−2.229%***	3.501	12
HAR-RV-GFSI	0.000%**	1.816	15	−2.802%**	1.804	18	−34.218%	−3.546	18
HAR-RV-GPR	−0.376%	−1.392	18	−0.141%	−1.266	15	−0.033%	−0.150	4
HAR-RV-CPR-L^{US_SM}	6.563%***	4.840	–	8.569%***	5.644	–	–	–	–
HAR-RV-CPR-O^{US_SM}	–	–	–	–	–	–	2.847%***	4.314	–

Table 12
Out-of-sample forecasting performance for predicting daytime oil realized volatility: Out-of-sample R-squared results

Models	one-day-horizon daytime oil realized volatility			five-day-horizon daytime oil realized volatility			22-day-horizon daytime oil realized volatility		
	R^2_{oos}	CW-stat	R^2_{oos} rank	R^2_{oos}	CW-stat	R^2_{oos} rank	R^2_{oos}	CW-stat	R^2_{oos} rank
HAR-RV-CPR	4.197%***	3.710	1	5.886%***	4.732	2	4.335%***	4.792	2
HAR-RV-L	−1.340%**	1.733	17	0.049%***	2.870	13	−3.445%	0.779	15
HAR-RV-J	−0.008%	0.262	14	0.112%	0.540	12	0.093%	0.722	4
HAR-RV-O	0.605%	1.047	10	0.927%**	2.056	9	−0.829%	−0.370	10
HAR-RV-RV^{CN_SM}	−0.835%	−1.272	16	0.155%	1.173	11	−0.840%	0.789	11
HAR-RV-L^{CN_SM}	0.019%	1.031	12	−0.137%	0.255	14	−0.401%	0.005	8
HAR-RV-J^{CN_SM}	0.023%	0.426	11	−0.784%	−1.904	16	−1.581%	−1.137	13
HAR-RV-O^{CN_SM}	0.659%*	1.457	9	0.942%**	2.027	8	−0.502%	0.103	9
HAR-RV-RV^{US_SM}	1.917%***	3.289	5	2.615%***	3.744	6	−3.247%*	1.325	14
HAR-RV-L^{US_SM}	4.035%***	4.460	2	6.229%***	4.972	1	−0.051%**	2.265	5
HAR-RV-J^{US_SM}	0.991%***	3.491	8	1.404%***	3.474	7	1.322%***	2.461	3
HAR-RV-O^{US_SM}	1.518%**	2.300	6	3.462%***	3.614	5	−0.088%	1.072	6
HAR-RV-IDEMV^{US_SM}	0.008%*	1.375	13	−3.294%**	1.849	17	−1.185%***	5.097	12
HAR-RV-VIX^{US}	3.266%***	3.816	3	3.598%***	5.058	4	−4.455%***	4.871	16
HAR-RV-OVX^{US}	1.074%***	2.373	7	0.493%**	2.287	10	−9.473%	−2.247	17
HAR-RV-GVX^{US}	2.002%***	2.353	4	5.181%***	4.687	3	9.537%***	5.503	1
HAR-RV-GFSI	−1.474%	1.240	18	−8.385%	0.708	18	−42.177%	−3.065	18
HAR-RV-GPR	−0.357%	−1.117	15	−0.220%	−1.042	15	−0.273%	−1.056	7
HAR-RV-CPR-L^{US_SM}	6.796%***	5.048	–	9.940%***	6.074	–	–	–	–
HAR-RV-CPR-GVX^{US}	–	–	–	–	–	–	12.984%***	6.241	–

et al., 2019). We regress the H-day-ahead average of abnormal trading volume on CPR and lagged abnormal trading volume[3]:

$$Z_{[t+1,t+H]} = \beta_0 + \beta_{CPR} CPR_t + \beta_{ABVol} ABVol_t + \varepsilon^H_{t+1}, \tag{9}$$

where $Z_{[t+1,t+H]} = \frac{1}{H}\sum_{h=1}^{H} ABVol_{t+h}$. Then, we regress the H-day-ahead ORV as follows:

$$Y_{[t+1,t+H]} = \beta_0 + \beta_Z Z_{[t+1,t+H]} + \beta_2 log(ORV_t) + \beta_3 log\left(ORV^w_t\right) + \beta_4 log\left(ORV^m_t\right) + \varepsilon^H_{t+1}, \tag{10}$$

where $Y_{[t+1,t+H]} = \frac{1}{H}\sum_{h=1}^{H} log(ORV_{t+h})$, and we use the full-day ORV to conduct our analysis. When fitting the regressions, we normalize the right-hand variables. We mainly focus on the coefficient of CPR in Eq. (9) and the coefficient of $Z_{[t+1,t+H]}$ in Eq. (10). In Table 14, CPR significantly and positively predicts future one- and five-day-ahead abnormal trading volume. This indicates that a high CPR increases market investor sentiment. Furthermore, in Columns 3 and 5, the results show that contemporaneous market investor sentiment significantly impacts market volatility, and the positive coefficients are in line with the intuition about the relationship between investor sentiment and price volatility. For the 22-day-ahead abnormal trading volume, CPR shows a positive coefficient but no significance. Combining these results of Eqs. (9) and (10) together, we show that the predictive role of CPR on ORV is partially related to the channel of market investor sentiment, at least in terms of predicting one- and five-day-ahead ORVs.

We examine the transmission mechanism from the liquidity perspective. The illiquidity measure of Amihud (2002) is used to capture

[3] We add the lagged abnormal trading volume in this regression to control the autocorrelation of the abnormal trading volume.

Table 13
Out-of-sample forecasting performance for predicting the nighttime oil realized volatility: Out-of-sample R-squared results

Models	one-day-horizon nighttime oil realized volatility			five-day-horizon nighttime oil realized volatility			22-day-horizon nighttime oil realized volatility		
	R^2_{oos}	CW-stat	R^2_{oos} rank	R^2_{oos}	CW-stat	R^2_{oos} rank	R^2_{oos}	CW-stat	R^2_{oos} rank
HAR-RV-CPR	1.812%**	2.045	5	2.597%***	2.652	5	4.868%***	3.721	2
HAR-RV-L	0.530%	1.997	9	−1.510%*	1.409	16	−1.415%	−0.267	14
HAR-RV-J	−0.345%**	1.216	14	0.797%**	2.161	8	0.213%	1.246	9
HAR-RV-O	−0.295%*	1.612	12	−3.022%	1.013	18	−2.245%	−0.597	15
HAR-RV-RV^{CN_SM}	0.628%**	2.168	8	0.648%**	2.038	9	−0.419%	−1.122	11
HAR-RV-L^{CN_SM}	−0.478%	−0.616	17	0.149%	0.894	10	0.830%**	1.786	7
HAR-RV-J^{CN_SM}	−0.179%	−1.119	11	−0.437%	−0.747	13	−2.619%	−1.531	17
HAR-RV-O^{CN_SM}	−0.083%	1.196	10	−0.081%*	1.455	12	0.262%	0.956	8
HAR-RV-RV^{US_SM}	1.687%***	2.879	6	3.934%***	3.846	4	1.402%**	2.145	5
HAR-RV-L^{US_SM}	3.272%***	3.085	3	2.540%***	2.910	6	0.940%**	2.110	6
HAR-RV-J^{US_SM}	−3.197%	−0.954	18	−1.145%	0.229	14	−0.676%	−1.485	13
HAR-RV-O^{US_SM}	1.062%**	2.197	7	1.577%**	2.120	7	1.491%**	2.223	4
HAR-RV-IDEMV^{US_SM}	−0.340%	1.140	13	−1.331%	0.064	15	−2.418%	−0.480	16
HAR-RV-VIX^{US}	3.878%***	3.544	1	8.806%***	5.213	1	6.459%***	5.236	1
HAR-RV-OVX^{US}	3.272%***	3.628	2	4.679%***	4.004	2	−0.586%	−1.701	12
HAR-RV-GVX^{US}	2.009%***	3.262	4	4.101%***	4.271	3	1.798%**	2.306	3
HAR-RV-GFSI	−0.405%*	1.488	16	−2.146%	0.925	17	−12.240%	−7.413	18
HAR-RV-GPR	−0.356%	−1.678	15	0.014%	0.260	11	−0.011%	0.199	10
HAR-RV-CPR-VIX^{US}	4.871%***	3.265	–	9.653%***	4.875	–	9.721%***	5.431	–

Table 14
Economic mechanism for the predictive role of CPR via investor sentiment perspective

	$Z_{[t+1,t+H]} = \frac{1}{H}\sum_{h=1}^{H} ABVol_{t+h}$			$Y_{[t+1,t+H]} = \frac{1}{H}\sum_{h=1}^{H} log(ORV_{t+h})$		
	H = 1		H = 5	H = 22		
	$Z_{[t+1,t+H]}$	$Y_{[t+1,t+H]}$	$Z_{[t+1,t+H]}$	$Y_{[t+1,t+H]}$	$Z_{[t+1,t+H]}$	$Y_{[t+1,t+H]}$
Intercept	1.011***	1.074***	1.009***	1.073***	1.007***	1.072***
t-stat	114.963	59.220	110.970	67.707	146.840	79.259
CPR_t	0.033***	–	0.017**	–	0.007	–
t-stat	3.350	–	2.037	–	0.907	–
$ABVol_t$	0.171***	–	0.108***	–	0.032***	–
t-stat	13.268	–	10.620	–	4.429	–
$Z_{[t+1,t+H]}$	–	0.394***		0.277***	–	0.306***
t-stat	–	17.819		15.669	–	21.186
$logORV_t$	–	0.096***		0.082***	–	0.050***
t-stat	–	3.186		4.118	–	3.239
$logORV_t^w$	–	0.217***		0.209***	–	0.120***
t-stat	–	5.957		7.086	–	5.118
$logORV_t^m$	–	0.186***		0.177***	–	0.202***
t-stat	–	6.390		7.189	–	11.589
R^2	29.431%	60.493%	21.875%	68.373%	4.982%	69.350%

Note: This table reports the results of the economic mechanism for the predictive role of CPR via the investor sentiment perspective. Here, the oil realized volatility ($logORV_t$) indicates full-day ORV. The t values of in-sample results are adjusted by the Newey and West (1987) method. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

market liquidity performance. Similar to the aforementioned analysis, we use Eqs. (9) and (10) and take the place of *ABVol* by *ILLIQ* and $Z_{[t+1,t+H]}$ by $\frac{1}{H}\sum_{h=1}^{H} ILLIQ_{t+h}$. $ILLIQ_t$ is the absolute value of oil returns at day t divided by the trading volume in millions. In Table 15, we find that CPR has no significant impact on future market illiquidity, including one-, five-, and 22-day-ahead illiquidity, even though these variables are indeed positively and significantly related to oil market volatility.

In short, the predictive role of CPR on ORV is partial via market investor sentiment instead of market liquidity.

4.2. Robustness check

4.2.1. The incremental predictive information of CPR based on the evidence from diffusion indices

We examine the predictive performance of diffusion indices extracted from multiple pieces of predictive information. Moreover, we answer whether the CPR indicator has incremental predictive information beyond the diffusion indices extracted from non-CPR predictors. Three diffusion methods are conducted to form diffusion indices, including principal component analysis (PCA), partial least squares (PLS), and scaled principal component analysis (SPCA) proposed by Huang et al.

Table 15
Economic mechanism for the predictive role of CPR via liquidity perspective

	$Z_{[t+1,t+H]} = \frac{1}{H}\sum_{h=1}^{H} ILLIQ_{t+h}$			$Y_{[t+1,t+H]} = \frac{1}{H}\sum_{h=1}^{H} log(ORV_{t+h})$		
	H = 1		H = 5	H = 22		
	$Z_{[t+1,t+H]}$	$Y_{[t+1,t+H]}$	$Z_{[t+1,t+H]}$	$Y_{[t+1,t+H]}$	$Z_{[t+1,t+H]}$	$Y_{[t+1,t+H]}$
Intercept	0.122***	1.074***	0.122***	1.073***	0.123***	1.072***
t-stat	19.144	53.294	22.801	56.827	28.051	57.548
CPR_t	0.003	–	0.003	–	0.002	–
t-stat	0.405	–	0.505	–	0.465	–
$ILLIQ_t$	0.055**	–	0.055***	–	0.049***	–
t-stat	2.457	–	3.050	–	3.173	–
$Z_{[t+1,t+H]}$	–	0.131***	–	0.093***	–	0.105***
t-stat	–	2.775	–	2.610	–	2.739
$logORV_t$	–	0.186***	–	0.125***	–	0.073***
t-stat	–	4.985	–	4.463	–	3.111
$logORV_t^w$	–	0.299***	–	0.281***	–	0.156***
t-stat	–	6.861	–	7.341	–	4.488
$logORV_t^m$	–	0.102***	–	0.118***	–	0.166***
t-stat	–	2.871	–	3.510	–	5.324
R^2	6.726%	42.931%	15.699%	54.133%	19.389%	41.889%

Note: This table reports the results of the economic mechanism for the predictive role of the CPR via the liquidity perspective. The t values of in-sample results are adjusted by the Newey and West (1987) method. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 16
Out-of-sample results of the predictive models using diffusion indices

Models	Forecasting the full-day oil realized volatility			Forecasting the daytime oil realized volatility			Forecasting the nighttime oil realized volatility		
	1-day	5-day	22-day	1-day	5-day	22-day	1-day	5-day	22-day
Panel A. The HAR-RV-CPR model									
HAR-RV-CPR	3.291%***	5.294%***	2.485%***	4.183%***	6.389%***	4.423%***	1.805%**	2.881%***	4.693%***
Panel B. The PLS diffusion index without CPR information									
HAR-RV-PLS$^{Non\text{-}CPR}$	4.212%***	4.076%***	−8.195%*	3.666%***	3.517%***	−8.987%***	5.214%***	8.342%***	2.268%***
HAR-RV-CPR-PLS$^{Non\text{-}CPR}$	**6.690%***	**7.844%***	−8.620%*	**7.126%***	**8.540%***	−10.420%***	**6.579%***	**10.679%***	**5.982%***
Panel C. The PCA diffusion index without CPR information									
HAR-RV-PCA$^{Non\text{-}CPR}$	4.351%***	5.222%***	−4.966%**	3.930%***	4.309%***	−6.223%***	5.114%***	9.557%***	4.848%***
HAR-RV-CPR-PCA$^{Non\text{-}CPR}$	**6.661%***	**8.563%***	−6.501%**	**7.269%***	**9.034%***	−8.591%***	**6.377%***	**11.735%***	**8.612%***
Panel D. The SPCA diffusion index without CPR information									
HAR-RV-SPCA$^{Non\text{-}CPR}$	2.978%***	3.040%***	−9.377%*	2.569%***	1.680%***	−11.039%***	4.210%***	6.778%***	0.556%*
HAR-RV-CPR-SPCA$^{Non\text{-}CPR}$	**5.665%***	**7.301%***	−8.598%**	**6.344%***	**7.505%***	−9.797%***	**5.415%***	**8.698%***	**4.483%***

(2021). The first principal component is taken as the diffusion indicator for each method, and we have the PCA$^{Non\text{-}CPR}$, PLS$^{Non\text{-}CPR}$, and SPCA$^{Non\text{-}CPR}$ indices. For out-of-sample forecasting, we obtain these diffusion indices step by step using the rolling samples.

In Table 16, we report the R^2_{OOS} of the models using diffusion indices. Panel A copies the results of the HAR-RV-CPR models. Panel B shows the results of the HAR-RV-X and HAR-RV-CPR-X models, where X is the PLS$^{Non\text{-}CPR}$ diffusion indicator. Basically, the R^2_{OOS} of the HAR-RV-CPR-PLS$^{Non\text{-}CPR}$ model is greater than that of the HAR-RV-PLS$^{Non\text{-}CPR}$ model, which is attributed to the effective incremental predictive information of CPR that is independent of other variables. The HAR-RV-CPR-PLS$^{Non\text{-}CPR}$ model also performs better than the HAR-RV-CPR model, excluding predicting the long-term full-day and daytime ORVs, because the diffusion index performs worse in long-term forecasting. In Panel C and Panel D, we show the results of predictive models using PCA$^{Non\text{-}CPR}$ and SPCA$^{Non\text{-}CPR}$, respectively. The findings are consistent with the results in Panel B. Among the three diffusion indices, the PCA$^{Non\text{-}CPR}$ indicator has a slight advantage in forecasting performance.

Overall, from the perspective of using multiple pieces of predictive information, we prove the effective incremental predictive information of CPR. Additionally, diffusion indices are reliable in predicting long-term full-day and daytime ORVs and are especially good at predicting nighttime ORVs. In particular, the diffusion indicator based on the PCA method is recommended.

4.2.2. MCS test

To further evaluate the effective role of predictive models, we conduct the MCS test for model priorities, where all predictive models include the benchmark HAR-RV model, HAR-RV-X models, and one HAR-RV-CPR-X model of interest. In Table 17, we show the predictive errors (MSE and MAE) and relative MCS p-values for predicting the full-day ORV. It is clear that the HAR-RV-CPR-L^{US_SM} model with the lowest

Table 17
Out-of-sample forecasting performance for predicting the full-day oil realized volatility: MCS test results

Models	one-day-horizon oil realized volatility				five-day-horizon oil realized volatility				22-day-horizon oil realized volatility			
	MSE	MCS p value	MAE	MCS p value	MSE	MCS p value	MAE	MCS p value	MSE	MCS p value	MAE	MCS p value
HAR-RV	0.425	0.000	0.488	0.000	0.204	0.000	0.342	0.000	0.159	0.053	0.298	0.009
HAR-RV-CPR	0.411	0.000	0.478	0.000	0.194	0.011	0.337	0.022	0.155	**0.147**	0.292	**1.000**
HAR-RV-L	0.423	0.001	0.490	0.000	0.207	0.001	0.345	0.000	0.168	0.000	0.307	0.001
HAR-RV-J	0.422	0.000	0.486	0.000	0.201	0.000	0.340	0.001	0.160	0.019	0.300	0.002
HAR-RV-O	0.419	0.000	0.487	0.000	0.203	0.001	0.341	0.003	0.160	0.033	0.299	0.006
HAR-RV-RV^{CN_SM}	0.423	0.000	0.485	0.000	0.203	0.000	0.341	0.000	0.160	0.027	0.300	0.003
HAR-RV-L^{CN_SM}	0.426	0.000	0.492	0.000	0.204	0.000	0.342	0.000	0.158	0.068	0.296	0.051
HAR-RV-J^{CN_SM}	0.426	0.000	0.488	0.000	0.204	0.000	0.343	0.000	0.161	0.003	0.301	0.000
HAR-RV-O^{CN_SM}	0.423	0.000	0.487	0.000	0.202	0.000	0.340	0.000	0.159	0.041	0.298	0.009
HAR-RV-RV^{US_SM}	0.417	0.000	0.484	0.000	0.199	0.001	0.340	0.000	0.167	0.002	0.311	0.000
HAR-RV-L^{US_SM}	0.405	0.000	0.474	0.000	0.192	0.002	0.335	0.006	0.159	0.041	0.302	0.000
HAR-RV-J^{US_SM}	0.423	0.000	0.486	0.000	0.202	0.000	0.343	0.000	0.159	0.009	0.299	0.002
HAR-RV-O^{US_SM}	0.409	0.002	0.482	0.000	0.196	0.001	0.336	0.022	0.158	0.065	0.298	0.009
HAR-RV-IDEMV^{US_SM}	0.421	0.000	0.485	0.000	0.204	0.009	0.342	0.011	0.163	**0.147**	0.293	**0.172**
HAR-RV-VIX^{US}	0.411	0.000	0.480	0.000	0.194	0.011	0.337	0.022	0.163	**0.147**	0.311	0.051
HAR-RV-OVX^{US}	0.422	0.000	0.486	0.000	0.205	0.001	0.344	0.000	0.185	0.002	0.329	0.005
HAR-RV-GVX^{US}	0.418	0.000	0.482	0.000	0.198	0.003	0.335	0.022	0.163	**0.147**	0.298	**0.117**
HAR-RV-GFSI	0.425	0.000	0.488	0.000	0.210	0.009	0.350	0.022	0.213	0.065	0.362	0.040
HAR-RV-GPR	0.427	0.000	0.489	0.000	0.204	0.000	0.342	0.000	0.159	0.053	0.298	0.009
HAR-RV-CPR-L^{US_SM}	0.397	**1.000**	0.466	**1.000**	0.187	**1.000**	0.331	**1.000**	–	–	–	–
HAR-RV-CPR-O^{US_SM}	–	–	–	–	–	–	–	–	0.155	**1.000**	0.292	**0.104**

Table 18
Out-of-sample forecasting performance for predicting the daytime oil realized volatility: MCS test results

Models	one-day-horizon daytime oil realized volatility				five-day-horizon daytime oil realized volatility				22-day-horizon daytime oil realized volatility			
	MSE	MCS p value	MAE	MCS p value	MSE	MCS p value	MAE	MCS p value	MSE	MCS p value	MAE	MCS p value
HAR-RV	0.343	0.000	0.439	0.001	0.152	0.000	0.296	0.000	0.123	0.000	0.273	0.000
HAR-RV-CPR	0.328	0.000	0.431	0.001	0.143	0.001	0.288	0.000	0.118	0.040	0.266	0.043
HAR-RV-L	0.347	0.009	0.429	0.036	0.151	0.001	0.297	0.001	0.127	0.002	0.277	0.001
HAR-RV-J	0.343	0.000	0.439	0.000	0.151	0.000	0.295	0.000	0.123	0.000	0.273	0.000
HAR-RV-O	0.341	0.000	0.434	0.021	0.150	0.000	0.295	0.001	0.124	0.001	0.274	0.000
HAR-RV-RV^{CN_SM}	0.345	0.000	0.440	0.000	0.151	0.000	0.296	0.000	0.124	0.004	0.274	0.004
HAR-RV-L^{CN_SM}	0.343	0.000	0.438	0.008	0.152	0.000	0.297	0.000	0.124	0.000	0.274	0.000
HAR-RV-J^{CN_SM}	0.343	0.000	0.439	0.001	0.153	0.000	0.298	0.000	0.125	0.002	0.276	0.000
HAR-RV-O^{CN_SM}	0.340	0.000	0.436	0.014	0.150	0.000	0.295	0.000	0.124	0.000	0.275	0.000
HAR-RV-RV^{US_SM}	0.336	0.000	0.434	0.009	0.148	0.000	0.289	0.000	0.127	0.008	0.282	0.006
HAR-RV-L^{US_SM}	0.329	0.000	0.430	0.010	0.142	0.012	0.285	0.025	0.123	0.010	0.275	0.008
HAR-RV-J^{US_SM}	0.339	0.000	0.435	0.014	0.149	0.000	0.293	0.000	0.122	0.010	0.272	0.003
HAR-RV-O^{US_SM}	0.337	0.000	0.436	0.002	0.146	0.000	0.291	0.001	0.123	0.000	0.274	0.000
HAR-RV-IDEMV^{US_SM}	0.343	0.000	0.439	0.001	0.157	0.000	0.300	0.000	0.125	0.001	0.279	0.003
HAR-RV-VIX^{US}	0.331	0.004	0.434	0.014	0.146	0.051	0.291	0.023	0.129	0.040	0.287	0.009
HAR-RV-OVX^{US}	0.339	0.004	0.437	0.029	0.151	0.036	0.292	0.025	0.135	0.010	0.288	0.008
HAR-RV-GVX^{US}	0.336	0.002	0.433	0.035	0.144	0.055	0.288	0.056	0.111	0.002	0.261	0.006
HAR-RV-GFSI	0.348	0.000	0.444	0.000	0.164	0.038	0.303	0.020	0.175	0.036	0.314	0.032
HAR-RV-GPR	0.344	0.000	0.440	0.001	0.152	0.000	0.297	0.000	0.124	0.000	0.273	0.000
HAR-RV-CPR-L^{US_SM}	0.319	**1.000**	0.423	**1.000**	0.136	**1.000**	0.281	**1.000**	–	–	–	–
HAR-RV-CPR-GVX^{US}	–	–	–	–	–	–	–	–	0.107	**1.000**	0.257	**1.000**

MSE and MAE is the only one belonging to the advantage model confidence set for the short- and middle-term full-day ORVs. The HAR-RV-CPR and HAR-RV-CPR-O^{US_SM} models are better at predicting the long-term full-day ORV.

The MCS results for predicting the daytime ORV are presented in Table 18. The findings are consistent with the R^2_{OOS} results. Specifically, the HAR-RV-CPR-L^{US_SM} model performs best for the short- and middle-term daytime ORVs, and the HAR-RV-CPR-GVX^{US} performs best for the long-term daytime ORV.

The MCS results for predicting the nighttime ORV are shown in Table 19. By comprehensively evaluating the MSE, MAE, and MSE- and MAE-based MCS p values, the HAR-RV-CPR-VIX^{US} model is still more reliable in predicting the short-, middle-, and long-term nighttime ORVs, as shown by the results of out-of-sample R-squared.

Thus, our findings are robust through evaluating predictive errors and examining MCS tests.

Table 19
Out-of-sample forecasting performance for predicting the nighttime oil realized volatility: MCS test results

Models	one-day-horizon nighttime oil realized volatility				five-day-horizon nighttime oil realized volatility				22-day-horizon nighttime oil realized volatility			
	MSE	MCS p value	MAE	MCS *p* value	MSE	MCS *p* value	MAE	MCS p value	MSE	MCS p value	MAE	MCS p value
HAR-RV	0.487	0.001	0.536	0.006	0.211	0.003	0.348	0.000	0.183	0.043	0.295	0.035
HAR-RV-CPR	0.478	0.001	0.534	0.023	0.206	0.006	0.347	0.001	0.174	0.043	0.291	**0.138**
HAR-RV-L	0.485	0.036	0.537	0.023	0.215	0.016	0.347	0.071	0.185	0.004	0.297	0.011
HAR-RV-J	0.489	0.000	0.538	0.000	0.210	0.001	0.347	0.000	0.182	0.043	0.296	0.046
HAR-RV-O	0.489	0.036	0.537	0.023	0.218	0.008	0.352	0.003	0.187	0.004	0.299	0.008
HAR-RV-RV^{CN_SM}	0.484	0.001	0.533	0.091	0.210	0.000	0.345	0.000	0.184	0.004	0.295	0.013
HAR-RV-L^{CN_SM}	0.489	0.000	0.537	0.003	0.211	0.001	0.347	0.000	0.181	0.043	0.293	0.099
HAR-RV-J^{CN_SM}	0.488	0.001	0.537	0.003	0.212	0.006	0.348	0.000	0.188	0.043	0.297	0.085
HAR-RV-O^{CN_SM}	0.488	0.002	0.537	0.003	0.212	0.008	0.348	0.000	0.182	0.043	0.294	0.077
HAR-RV-RV^{US_SM}	0.479	0.001	0.533	0.023	0.203	0.016	0.346	0.054	0.180	0.043	0.302	0.085
HAR-RV-L^{US_SM}	0.471	0.047	0.528	**0.134**	0.206	0.008	0.348	0.014	0.181	0.029	0.296	0.085
HAR-RV-J^{US_SM}	0.503	0.002	0.542	0.000	0.214	0.001	0.353	0.000	0.184	0.012	0.297	0.018
HAR-RV-O^{US_SM}	0.482	0.036	0.538	0.005	0.208	0.016	0.347	0.001	0.180	0.043	0.294	0.085
HAR-RV-$IDEMV^{US_SM}$	0.489	0.035	0.540	0.002	0.214	0.016	0.347	0.105	0.187	0.043	0.292	**0.138**
HAR-RV-VIX^{US}	0.468	0.047	0.529	**0.126**	0.193	0.054	0.337	**1.000**	0.171	0.043	0.292	0.085
HAR-RV-OVX^{US}	0.471	0.047	0.529	**0.134**	0.202	0.016	0.342	**0.105**	0.184	0.026	0.296	0.006
HAR-RV-GVX^{US}	0.477	0.002	0.532	0.091	0.203	0.016	0.340	**0.105**	0.180	0.043	0.289	**0.138**
HAR-RV-GFSI	0.489	0.001	0.538	0.017	0.216	0.003	0.352	0.034	0.205	0.005	0.314	0.004
HAR-RV-GPR	0.489	0.001	0.538	0.001	0.211	0.002	0.348	0.000	0.183	0.032	0.295	0.035
HAR-RV-CPR-VIX^{US}	0.463	**1.000**	0.527	**1.000**	0.191	**1.000**	0.338	**0.105**	0.165	**1.000**	0.289	**1.000**

Note: The values in bold indicate that the corresponding model belongs to the advantage MCS. The values in bold and underlined indicate that the corresponding model has the highest MCS p value. The entire sample period is from 26 March 2018, to 30 June 2022. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

5. Conclusion

In this study, we mainly examine the effective predictive role of CPR in the Chinese ORV. Based on our in-sample and out-of-sample evidence, we show that CPR has incremental predictive information that is independent of other conventional domestic and foreign predictors. In particular, we also find that many predictors from the US stock and options markets, such as the stock market leverage factor, VIX, and gold implied volatility, have a more important predictive role for the Chinese ORV than the predictors from domestic stock and oil markets. Additionally, considering the potential difference in investors participating during the daytime and nighttime trading periods in China, we also forecast the daytime and nighttime ORVs. The evidence shows that the CPR indicator effectively predicts both daytime and nighttime ORVs; in particular, the VIX is a very important predictor of nighttime realized volatility in the short, middle, and long term.

Our research alerts that the Chinese government and academia need to focus on the influencing factors and predictability of the nighttime ORV for the domestic oil futures market despite the relatively high predictability of the full-day ORV. Policymakers can well predict future oil market risks based on CPR, negative returns of the US stock market, and the VIX and gold implied volatility of the US options market, among others, and then develop risk prevention plans to avoid potential market risk. Additionally, domestic and foreign investors can benefit from our study to adjust the weights of their oil assets in portfolios and design new investment strategies and risk aversion procedures, especially considering the different driving factors for daytime and nighttime oil price volatilities.

According to our study and previous investigations, CPR shows an excellent predictive role in both asset returns and volatilities in the Chinese financial markets. The belief-based momentum indicator is promising; in future works, we will investigate the role of CPR in returns of the Chinese/global oil futures market that directly examines the "price momentum effect" of CPR.

Credit author statement

All authors equally contribute to this research.

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Appendix A. Supplementary data

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